



An intelligent multi-objective decision framework for geothermal–LNG assisted co-generation of power, natural gas, and hydrogen

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ABSTRACT

This study presents an intelligent framework for optimizing geothermal–LNG assisted tri-generation systems, integrating machine learning, Pareto-based optimization, and scenario-based decision-making to enhance performance and operational decision-making. The system simultaneously produces electricity, hydrogen, and natural gas while recovering the cold energy of liquefied natural gas (LNG), utilizing a dual-well geothermal source, an ORC, and a PEM electrolyzer. Six key decision variables including Separator 1 and 2 pressures (P_{S1} , P_{S2}), ORC evaporator temperature (T_{EVP}), LNG pump pressure (P_{LNG}), energy allocation to electrolysis (α), and PEM temperature (T_{PEM}), were analyzed with respect to exergy efficiency (EE), power-specific cost (PSC), and hydrogen production rate (MFR_{H_2}). Hybrid ANN models optimized via PSO, MPA, and WSO accurately predicted system outputs, with PSO-ANN achieving the highest accuracy for EE (MAPE < 0.03%), MPA-ANN for PSC (MAPE < 0.02%), and WSO-ANN for hydrogen production (MAPE < 0.3%). Multi-objective material generation algorithm (MOMGA) produced a well-distributed Pareto front, revealing trade-offs: increasing EE generally raised PSC, while maximizing hydrogen reduced EE. Key operational windows were identified (P_{S1} : 603–620 kPa, T_{EVP} : 162–164 °C, α : 0.25–0.28), ensuring balanced efficiency, cost, and hydrogen yield. Weighted aggregated sum product assessment (WASPAS) enabled flexible multi-criteria decision-making, allowing multi-objective prioritization. The proposed framework demonstrates robust predictive capability, systematic exploration of trade-offs, and actionable operational guidance, offering a comprehensive tool for the design and deployment of integrated geothermal–LNG tri-generation systems with enhanced techno-economic and hydrogen production performance.

1. Introduction

The accelerating expansion of global energy consumption, alongside strict environmental regulations and depletion concerns associated with conventional fuels, has intensified the search for both sustainable power

resources and advanced methods of energy recovery [1]. Traditional energy infrastructures often overlook substantial exergy losses, particularly in systems where temperature differentials remain unexploited or where renewable heat sources are not adequately integrated with high-grade cold resources [2]. In this regard, geothermal reservoirs,

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characterized by their consistent thermal output and independence from weather fluctuations, have emerged as an attractive renewable alternative capable of delivering dependable base-load energy [3]. At the same time, large-scale handling of liquefied natural gas (LNG) continues to generate considerable cryogenic potential during regasification, a valuable stream of cold energy that is commonly dissipated into the environment without any economic return [4]. The coexistence of these two unconventional resources (one offering stable heat and the other providing ultra-low-temperature exergy) suggests a significant opportunity to develop hybrid conversion systems that minimize waste while expanding energy pathways [5].

Recent advances in thermochemical and power-generation technologies provide a mechanism for such integration, allowing geothermal heat to be upgraded through simultaneous use of cryogenic cooling to drive high-efficiency conversion cycles [6]. This synergy creates a platform not only for electricity generation but also for the upgrading of natural gas and the production of hydrogen through enhanced heat-to-fuel conversion [7]. As global interest in hydrogen accelerates, driven by decarbonization strategies and the rapid growth of clean-fuel markets, designing systems capable of co-producing gas, power, and hydrogen in a flexible manner becomes technologically appealing [8]. Such multi-output architectures can exploit varying exergy streams to support diversified energy portfolios, improve system resilience, and reduce operational costs through strategic energy coupling [9]. Consequently, the convergence of geothermal resources with LNG cold recovery presents a novel direction for future energy systems, enabling more efficient, intelligent, and sustainable pathways for integrated fuel and power production.

Several researchers have investigated the integration of geothermal energy with multi-generation systems to simultaneously produce electricity, hydrogen, cooling, heating, and freshwater, highlighting the potential of combining novel cycles and energy recovery strategies to enhance system efficiency. For example, Mehdikhani et al. [10] developed a hybrid geothermal-based plant incorporating a flash system, organic Rankine cycle (ORC), LNG regasification, and proton exchange membrane (PEM) electrolysis using two wells with different thermodynamic conditions. The system achieved 13,490 kW net power, 3.575 kg/h hydrogen, 17.14% thermal efficiency, and 44.16% exergy efficiency, with specific power cost of 19.86 \$/GJ, demonstrating the sensitivity of power output and efficiencies to high-temperature well conditions.

Building on the use of LNG cold energy recovery, Lin et al. [11] proposed a double-flash geothermal system integrated with Kalina and ammonia Rankine cycles and PEM electrolysis. Their system produced 30,640 kW electricity, 15,260 kW cooling, and 20.16 kg/h hydrogen, achieving 28.91% energy efficiency and 76.39% exergy efficiency, while reducing CO₂ emissions by 189,898 kg/year. The study highlighted the LNG evaporator as the major source of irreversibility (68.1%), suggesting significant optimization potential. Similarly, Emadi et al. [12] examined a system combining Kalina, Stirling, and LNG cycles with a PEM electrolyzer, producing 10.69 MW net power, 6.09 MW cooling, and 204.77 kg/h hydrogen, with 43% exergy efficiency, emphasizing the influence of key input parameters on performance.

Ozturk and Dincer [13] investigated a flash-binary system with an alkaline electrolyzer for hydrogen blending into natural gas, achieving energy and exergy efficiencies of 46.8% and 77.9%, respectively. Seasonal exergy efficiency ranged from 76.98% to 86.63%, and increased hydrogen content reduced CO₂ and heat release while enhancing oxygen output. Extending hybrid configurations, Manesh et al. [14] proposed a solar-geothermal-natural gas polygeneration plant integrating ORC, internal combustion engine, PEM electrolyzer, and humidification-dehumidification desalination, producing 4.67 m³/day freshwater, 1.85 kg/h hydrogen, and 1.31 kg/s hot water, with energy and exergy efficiencies of 23.87% and 28.21%.

Further studies explored multi-objective thermodynamic and economic optimization. Tjahjono et al. [15] developed an LNG-driven

system combining CO₂ and ORC cycles with reverse osmosis and NaClO production, achieving 54.3% energy efficiency and 13.1% exergy efficiency, along with annual production of 10.53 GWh electricity, 276.4 GWh cooling, 17,280 m³ potable water, and 739.56 tons hydrogen. Xie and Zheng [16] optimized a trans-critical CO₂ Rankine-LNG system with ejector cooling and desalination, producing 435.9 kW electricity, 251.5 kW cooling, 3.68 kg/h hydrogen, and 0.0493 kg/s freshwater with 16.99% exergy efficiency and a payback period of 0.616 years.

Recent works have further integrated Kalina cycles, double-effect absorption chillers, and ORC systems with LNG regasification. Celik-Toker et al. [17] achieved 2273 kW electricity, 6320 kW district heating, 1.784 kg/h hydrogen, and 7.51 m³/h freshwater, with energy and exergy efficiencies of 0.73 and 0.32. Zhang et al. [18] reported energy and exergy efficiencies of 52.79% and 85.22% in a similar CCHP system, with highest irreversibility in LNG regasification (81%) and a total cost rate of 186.74 \$/h. Finally, Ata [19] applied multi-objective optimization using the response surface method (RSM), enhancing exergy efficiency by 2.19%, reducing payback period by 22.44%, decreasing CO₂ footprint by 1.37 kg/kWh, and improving net present value and levelized multi-energy cost, achieving a high desirability score of 0.8584. Overall, these studies collectively demonstrate that integrating geothermal energy with advanced multi-generation systems, LNG cold recovery, and hybrid cycles can significantly enhance energy, exergy, and economic performance while enabling the simultaneous production of power, cooling, freshwater, and hydrogen.

Recently, machine learning (ML) and artificial intelligence (AI) play a pivotal role in the development and operation of an integrated geothermal-LNG energy system by enabling accurate modeling, dynamic optimization, and intelligent decision-making across highly nonlinear thermodynamic interactions [20]. Such hybrid systems involve multi-physics behavior, varying exergy levels, and dynamic resource availability, which are difficult to predict using deterministic or conventional analytical tools [21]. AI techniques enhance the prediction of geothermal reservoir performance, assess cryogenic availability in LNG regasification, and estimate hydrogen yield under complex operating conditions, enabling reliable performance evaluation without extensive experimental data [22]. Moreover, machine-learning-driven multi-objective optimization supports simultaneous enhancement of power output, fuel upgrading, hydrogen production, and energy cost minimization by exploring vast design spaces efficiently [23]. Therefore, integrating AI-based modeling and optimization has become essential in advancing multigeneration systems, motivating a critical review of existing research trends and technological gaps in geothermal-LNG assisted energy frameworks.

Numerous studies have examined the benefits of hybrid system designs and the application of AI and ML for performance prediction, modeling, and multi-objective optimization of geothermal-based multi-generation systems for the simultaneous production of electricity, hydrogen, cooling, and freshwater. For instance, Shakibi et al. [24] developed a geothermal-based system for Australia, achieving 3946 kW net power, 152 kg/h freshwater, 8.73 kg/h hydrogen, and an exergy efficiency of 46.27%, with a net present value of 19.52 M\$ using artificial neural network (ANN)-based optimization. Similarly, Javaherian et al. [25] proposed a dual-source hybrid plant integrating gas turbine, supercritical CO₂ recompression Brayton cycle, ORC, absorption cooling, and reverse osmosis desalination, achieving 288.2 MW net power with 45% energy efficiency, 54% exergy efficiency, and a 7.2-year payback, while ANN-assisted optimization combined with genetic and grey wolf algorithms significantly reduced CO₂ emissions to 110.1 kg/MWh. Xu et al. [26] examined a waste-to-energy system combining municipal solid waste gasification with a proton-conducting solid oxide fuel cell, achieving 323.6 kW power, 854.2 g/s heat output, 40.3% electrical efficiency, 36.1% exergy efficiency, and 893.5 g/kWs emissions, with machine learning models predicting performance with R² values above 94%.

Chitgar et al. [27] developed a geothermal multi-generation system

and used artificial neural network–genetic algorithm optimization, showing that optimal working-fluid combinations can maximize electrical efficiency, hydrogen, and water production while minimizing cost. Li et al. [28] proposed a geothermal tri-generation system integrating a modified Kalina cycle, ORC, LNG subsystem, ejector cooling, LiBr–H₂O absorption refrigeration, and PEM electrolyzer. Using a hybrid Genetic–Fgoalattain algorithm for tri-objective optimization, the system produced 451.8 kW power, 297.8 kW cooling, 2.274 kg/h hydrogen, with an exergy efficiency of 26.5% and a payback period of 2.385 years. Faramarzi and Khavari [29] developed a geothermal-powered hydrogen liquefaction system coupled with LNG production, producing 101,100 kg liquid hydrogen and 63 kg LNG, reducing specific energy consumption by 25% and cost by 6% through genetic algorithm optimization.

Integration of LNG cold energy and dual-loop ORCs has been widely investigated. Li et al. [30] proposed a dual-loop ORC–LNG system, achieving a maximum net output power of 294.1 kW and thermal efficiency of 34.2%, with avoidable exergy destruction reduced from 38.2% to 23.4% through NSGA-II and entropy-weighted TOPSIS optimization. Mehrenjani et al. [31] optimized two geothermal-based multigeneration systems using ANNs and genetic algorithms, achieving 32,488 m³/day desalination in System 1 and 24.65 MW cooling. Almutairi [32] developed a dual-loop ORC integrated with LNG regasification and thermoelectric generation, producing 5.15 MW electricity, 127.31 m³/h desalinated water, and hydrogen while optimizing system performance using ANNs. Also, Asadabadi et al. [33] presented a hybrid solar-geothermal system with high-temperature cascaded ORC and LNG stream integration, achieving 14.73 MW net power, 132.02 kg/h liquid hydrogen, 770.28 kg/s freshwater, and 7.65 MW cooling with 29.69% exergy efficiency through ANN and genetic algorithm-based optimization.

Recent studies have explored hybrid biomass–LNG [34] and biogas–LNG [35] multigeneration systems using advanced 4E–5E analyses and AI-based optimization frameworks. However, none have examined a geothermal–LNG-driven tri-generation configuration that simultaneously integrates hydrogen production and LNG cold-energy recovery within a unified exergy–economic optimization environment. The present work addresses this gap by thermodynamically coupling geothermal heat utilization with LNG regasification and optimizing the integrated system through a structured, multi-scenario framework. This approach extends conventional 4E/5E + AI methodologies by offering actionable operational guidance tailored to geothermal–LNG energy hubs. Overall, the study advances the state of the art by introducing an intelligent, multi-layered framework that combines surrogate modeling, Pareto-based multi-objective search, and scenario-driven decision-making to enhance both system performance and practical design choices. In the machine learning component, hybrid models integrating multilayer perceptron neural networks (MLPNN) with advanced metaheuristic optimizers (including particle swarm optimization (PSO–MLPNN), marine predators algorithm (MPA–MLPNN), and white shark optimizer (WSO–MLPNN)) are employed to accurately predict system outputs under highly nonlinear thermodynamic interactions. For optimization, the study applies a multi-objective material generation algorithm (MOMGA) to explore trade-offs between exergy efficiency, power-specific cost, and hydrogen production, generating a robust set of Pareto-optimal solutions. Finally, the weighted aggregated sum product assessment (WASPAS) method is utilized to systematically rank these solutions, providing a comprehensive decision-making framework to identify the most balanced and practically implementable configuration. The overall workflow of this AI-based integrated framework is illustrated in Fig. 1, demonstrating how predictive modeling, optimization, and decision analysis are seamlessly coupled to guide the design of highly efficient, multi-output geothermal–LNG energy systems.

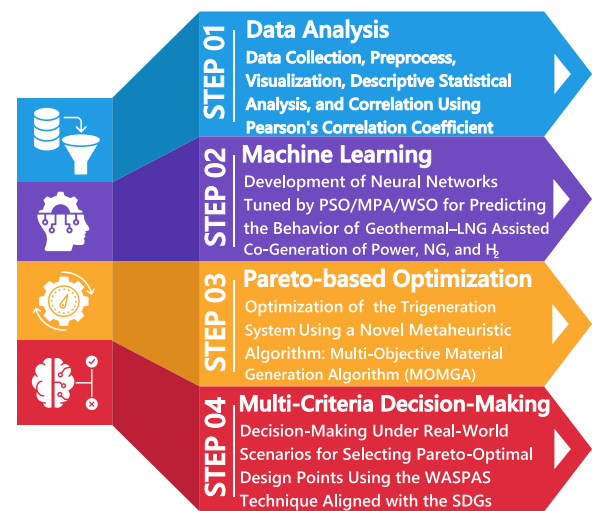


Fig. 1. Schematic workflow of the proposed intelligent framework integrating ML-based prediction, Pareto-based optimization, and scenario-based decision-making for geothermal–LNG assisted co-generation of electricity, natural gas, and hydrogen.

2. Data analysis

2.1. Configuration of the trigeneration system

The data reported by Naderi et al. [36] were used as the foundational basis for developing the novel framework proposed in the present research. Their work focused on the optimal design of a geothermal-based tri-generation system that simultaneously produced electricity, natural gas, and hydrogen while recovering the cold energy of LNG. Using a dual-well geothermal source, they aimed to improve the system's overall exergy efficiency, reduce the power-specific cost, and increase the hydrogen production rate. To achieve this goal, they applied a classical analysis based on RSM. Their research demonstrated the economic and thermodynamic potential of geothermal energy integration when complemented by cold-energy recovery and hydrogen generation using a PEM electrolyzer.

Naderi et al. generated the operating points required for thermodynamic simulations using a Box–Behnken Design (BBD), which is known for its efficiency and balanced exploration of multidimensional design spaces. BBD systematically assigns experimental runs at three coded levels (−1, 0, +1) for each decision variable, enabling the detection of nonlinearities and interaction effects while avoiding extreme corner points that may produce non-physical or unstable operating states. This characteristic makes BBD particularly suitable for complex energy systems, where boundary conditions may violate thermodynamic or equipment constraints. By applying BBD to the selected decision variables, a structured and orthogonal dataset was generated, ensuring uniform coverage of the feasible region with a minimal number of simulations. This design strategy enhances computational efficiency while preserving the fidelity and representativeness of the sampled operating space. In addition, an analysis of variance (ANOVA) was performed to evaluate the statistical validity of the RSM-based models and to confirm the significance of nonlinear interactions among the parameters.

In their thermodynamic configuration, the integrated plant consisted of four main cycles, each assigned a distinct operational role: the geothermal cycle, the LNG cycle, ORC, and the PEM electrolyzer cycle. As illustrated in Fig. 2, the system utilized geothermal fluid extracted from two wells with different thermal conditions. The fluid from Geothermal Well 1 (GW1) was flashed in the first separator, sending vapor to the high-pressure turbine and liquid to a second separator for further steam extraction. In parallel, the fluid from Geothermal Well 2

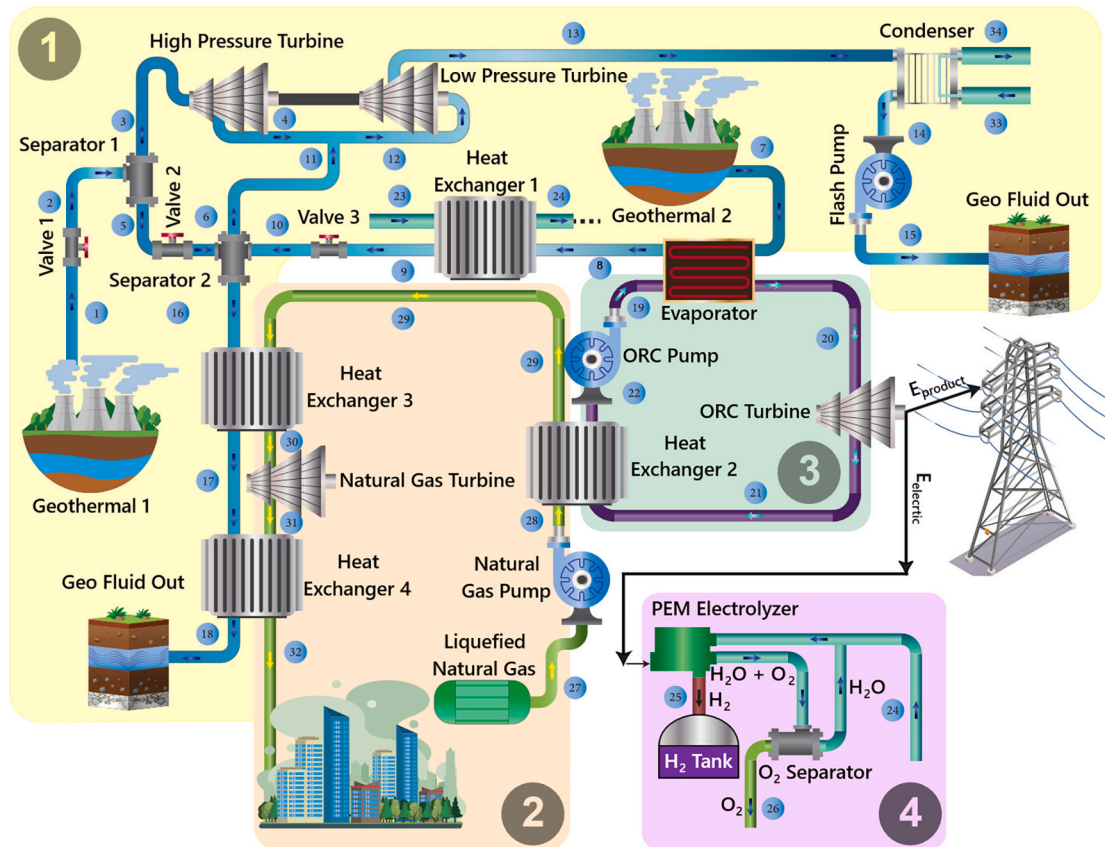


Fig. 2. Schematic configuration of the integrated geothermal-based tri-generation system, illustrating the four coupled cycles: (1) Geothermal, (2) LNG, (3) ORC, and (4) PEM electrolyzer.

supplied the ORC evaporator, where the organic working fluid was heated, pressurized, expanded in the ORC turbine, condensed, and recirculated via an ORC pump. The high-temperature effluent leaving the ORC evaporator subsequently heated feedwater to the optimal temperature (80 °C) required for electrolysis in the PEM unit. Inside the electrolyzer, water dissociated into hydrogen at the cathode and an oxygen-rich water mixture at the anode. Meanwhile, LNG was pressurized, superheated through a sequence of heat exchangers, expanded in the natural gas turbine, and further warmed before distribution to the end-user network. Thus, each background-colored block in Fig. 2 represented a complete thermodynamic cycle: (1) Geothermal, (2) LNG, (3) ORC, and (4) PEM, collectively forming a fully integrated trigeneration configuration.

Within this integrated configuration, six key decision variables govern the system performance: Separator 1 pressure (P_{S1}), Separator 2 pressure (P_{S2}), ORC evaporator temperature (T_{EVP}), LNG pump pressure (P_{LNG}), energy-allocation ratio to electrolysis (α), and PEM operating temperature (T_{PEM}). These variables were optimized with respect to three target outputs: exergy efficiency (EE), power-specific cost (PSC), and hydrogen production rate (MFR_{H2}).

The thermodynamic modeling of the system is carried out using Engineering Equation Solver (EES), which provides an extensive database of thermophysical properties and enables accurate evaluation of energy and exergy flows within the integrated system. The simulation is performed under steady-state conditions and is based on several commonly adopted assumptions. All components operate in steady-state mode, while pressure drops in pipelines and heat losses from heat exchangers are neglected. The isentropic efficiencies of turbines and pumps are assumed to be 85% and 80%, respectively. The environmental reference state used in the exergy calculations corresponds to an ambient temperature of 25 °C and a pressure of 101.3 kPa. In the ORC

subsystem, R141b is selected as the working fluid because of its favorable thermodynamic properties, moderate critical temperature of approximately 206.8 °C, critical pressure of 4.46 MPa, normal boiling point of about 32 °C, and low environmental impact indicators (ODP $\approx$ 0 and GWP $\approx$ 0.12). The ORC evaporator pinch-point temperature difference is considered within the range of 5–15 °C to ensure feasible heat transfer performance.

The thermodynamic behavior of the integrated system is governed by the fundamental principles of mass and energy conservation. For each component, the mass balance requires that the sum of inlet mass flow rates equals the sum of outlet mass flow rates, which can be expressed as $\sum \dot{m}_{in} = \sum \dot{m}_{out}$, where $\dot{m}$ represents the mass flow rate of the working fluid. Similarly, the first law of thermodynamics is applied to each control volume such that the difference between heat transfer rate and work transfer rate equals the difference between the total enthalpy flow rates entering and leaving the component, which is written in the form $Q - W = \sum \dot{m}_{out} h_{out} - \sum \dot{m}_{in} h_{in}$, where h denotes specific enthalpy. The performance of pumps and turbines is evaluated using isentropic efficiency relations. For a pump, the efficiency is defined as $\eta_p = v(P_{out} - P_{in}) / (h_{out} - h_{in})$, where v is the specific volume of the fluid and P_{in} and P_{out} are the inlet and outlet pressures. For turbines, the isentropic efficiency is expressed as $\eta_t = (h_{in} - h_{out}) / (h_{in} - h_{out,s})$, where $h_{out,s}$ is the outlet enthalpy corresponding to an ideal isentropic expansion process.

In the ORC subsystem, the heat transfer between the geothermal fluid and the organic working fluid is governed by the evaporator energy balance. The thermal energy released by the geothermal fluid, calculated from the difference between its inlet and outlet enthalpies, is equal to the energy gained by the ORC working fluid during evaporation. This balance can be written as $\dot{m}_{gf}(h_{gf,in} - h_{gf,out}) = \dot{m}_{ORC}(h_{evap,out} - h_{evap,in})$, where $\dot{m}_{gf}$ and $\dot{m}_{ORC}$ denote the mass flow rates of geothermal fluid and

ORC working fluid, respectively. The evaporator operation is further constrained by the pinch-point temperature difference to ensure feasible heat transfer between the two streams.

Hydrogen production is accomplished using a proton exchange membrane electrolyzer that converts electrical energy into chemical energy through water electrolysis. In this process, water is dissociated into hydrogen at the cathode and oxygen at the anode. The total energy required for this electrochemical reaction is determined by the thermodynamic relation $\Delta H = \Delta G + T\Delta S$, where ΔH is the enthalpy change of the reaction, ΔG represents the Gibbs free energy, and $T\Delta S$ corresponds to the thermal contribution associated with entropy change. The electrical power required for hydrogen production is given by $E_{electric} = J V$, where J denotes the current density and V represents the electrolyzer cell voltage. The total cell voltage is composed of several contributions including the reversible voltage V_o , the ohmic overpotential V_{ohm} caused by membrane resistance, and the activation overpotentials at the anode and cathode electrodes, denoted as $V_{act,a}$ and $V_{act,c}$. Consequently, the total cell voltage can be expressed as $V = V_o + V_{ohm} + V_{act,a} + V_{act,c}$, which reflects the different electrochemical losses occurring in the PEM electrolyzer.

To evaluate the thermodynamic performance of the integrated system, an exergy analysis is performed considering both physical and chemical exergy contributions. The total exergy rate associated with a stream is expressed as the sum of physical and chemical exergy components, written as $\dot{E}_{x_{tot}} = \dot{E}_{x_{ph}} + \dot{E}_{x_{ch}}$. The physical exergy of a stream is calculated from the relation $\dot{E}_{x_{ph}} = \dot{m}[(h - h_o) - T_o(s - s_o)]$, where h and s denote the specific enthalpy and entropy at the actual state, while h_o and s_o correspond to the thermodynamic properties at the reference environment defined by $T_o = 25^\circ\text{C}$ and $P_o = 101.3\text{ kPa}$. Exergy destruction within each component represents the thermodynamic irreversibility of the process and is calculated as $\dot{E}_{x_D} = \dot{E}_{x_F} - \dot{E}_{x_P} - \dot{E}_{x_L}$, where $\dot{E}_{x_F}$ denotes fuel exergy, $\dot{E}_{x_P}$ represents product exergy, and $\dot{E}_{x_L}$ accounts for exergy losses.

In order to incorporate economic considerations into the thermodynamic analysis, an exergoeconomic evaluation is performed. The cost balance equation for the k th component relates the cost rates of incoming and outgoing exergy streams with the capital investment cost and operating costs of the component. This balance can be expressed as $\dot{C}_{q,k} + \sum \dot{C}_{in,k} + \dot{Z}_k = \sum \dot{C}_{out,k} + \dot{C}_{w,k}$, where $\dot{C}_{in,k}$ and $\dot{C}_{out,k}$ represent the cost rates associated with inlet and outlet streams, $\dot{C}_{q,k}$ corresponds to the cost rate of heat transfer, and $\dot{C}_{w,k}$ denotes the cost rate associated with work interactions. The cost rate associated with a given exergy stream is calculated using the relation $\dot{C}_k = c_k \dot{E}_{x_k}$, where c_k represents the unit cost per unit exergy. In addition, several auxiliary exergoeconomic indicators are defined including the unit exergy cost of fuel $c_{F,k} = \dot{C}_{F,k}/\dot{E}_{x_{F,k}}$, the unit cost of product $c_{P,k} = \dot{C}_{P,k}/\dot{E}_{x_{P,k}}$, the exergoeconomic factor $f_k = \dot{Z}_k/(\dot{C}_{L,k} + \dot{C}_{D,k} + \dot{Z}_k)$, and the cost rate associated with exergy destruction $\dot{C}_{D,k} = c_{F,k}\dot{E}_{x_{D,k}}$.

The annualized capital cost rate of each component is determined using the capital recovery factor method. The cost rate $\dot{Z}_k$ is obtained from $\dot{Z}_k = (CRF \times \varphi / (N \times 3600)) Z_k$, where Z_k represents the investment cost of the component, φ is the maintenance factor, and N denotes the annual operating time of the system. The capital recovery factor is calculated as $CRF = [i_r(1 + i_r)^n] / [(1 + i_r)^n - 1]$, where i_r is the interest rate and n represents the system lifetime. In the present study, the annual operating time is considered to be 7446 h, the maintenance factor is taken as 1.06, the interest rate is assumed to be 10%, and the economic lifetime of the system is considered as 20 years.

The economic model further incorporates component-specific cost correlations for the major equipment used in the system. For example, the investment cost of geothermal separators is estimated as $Z = 1.218(42 + 1.63\dot{m}/3.6)/1000$, where $\dot{m}$ denotes the mass flow rate. The costs of turbines and pumps are expressed as functions of their power

capacity; for instance, the flash pump cost is calculated as $Z = 3540(\dot{W}_p)^{0.71}$ and the geothermal turbine cost is estimated as $Z = 6000(\dot{W}_t)^{0.7}$. Heat exchanger costs depend on the heat transfer area and follow the relation $Z = 2143A^{0.514}$, where A is the heat transfer surface area. The PEM electrolyzer investment cost is estimated from the electrical power requirement using $Z = 1000E_{electric}$, while the LNG pump cost is expressed as $Z = 1120(\dot{W}_p)^{0.8}$. These cost relations allow the evaluation of the economic contribution of each subsystem within the integrated trigeneration configuration.

The overall thermodynamic performance of the system is quantified using several key indicators. The net power output of the system is calculated as $\dot{W}_{net} = \dot{W}_{HPT} + \dot{W}_{LPT} + \dot{W}_{NGT} + \dot{W}_{ORCT} - (\sum \dot{W}_{pump} - \dot{E}_{electric})$, where the terms correspond to the power generated by the high-pressure turbine, low-pressure turbine, natural gas turbine, and ORC turbine, minus the electrical consumption of pumps and the PEM electrolyzer. The exergy efficiency of the integrated system, which represents the ratio between useful exergy output and total exergy input, is defined as $\eta_{ex} = \frac{\dot{E}_{x_{product}}}{\dot{E}_{x_{fuel}}} = \frac{\dot{W}_{net} + \dot{E}_{x_{25}} + \dot{E}_{x_{32}}}{\dot{E}_{x_1} + \dot{E}_{x_7} + \dot{E}_{x_{27}}}$. The economic

performance of the system is evaluated using the power-specific cost indicator PSC, which represents the cost per unit of useful energy output and is obtained by dividing the total cost rate of the system by the useful energy production rate. Finally, the hydrogen production performance is characterized by the hydrogen mass flow rate MFR_{H_2} , which represents the mass of hydrogen generated per unit time by the PEM electrolyzer and is directly related to the electrical power supplied to the electrolyzer and the electrochemical conversion efficiency.

In the LNG subsystem, natural gas is produced through LNG pressurization, vaporization, and expansion, enabling simultaneous cold-energy recovery and power generation. The NG mass flow rate is not treated as an independent decision variable; instead, it is determined by the thermodynamic behavior of the LNG cycle and governed by the mass and energy balance equations implemented in the EES model. As a result, NG delivery is inherently ensured by the system configuration and operating constraints. Since its variation within the feasible operating range is relatively limited, NG output was not considered as a separate optimization objective. Its contribution is implicitly reflected in the evaluation of exergy efficiency and power-specific cost through the overall energy and exergy balances of the system.

While traditional RSM can generate predictive expressions and offer insight into how design variations influence performance metrics, its reliance on fixed polynomial forms restricts its capacity to represent strongly nonlinear and tightly interdependent phenomena with sufficient fidelity. As a result, the standard RSM formulation often struggles to achieve reliable accuracy or broad applicability when confronted with intricate variable interactions. The framework introduced in this study adopts advanced machine learning models combined with meta-heuristic optimization techniques, which are inherently suited to identifying and reproducing complex nonlinear behaviors with superior precision. Accordingly, the dataset reported by Naderi et al. forms the basis of an AI-driven optimization framework that delivers superior accuracy, robustness, and computational efficiency compared with RSM.

2.2. Data description and visualization

Table 1 summarizes the descriptive statistics of the six operational inputs and three performance targets used in the tri-generation system, providing key insights into their variability, central tendency, and distribution behavior. Among the inputs, P_{S1} ranged from 600 to 1000 kPa, with a mean of 810 kPa and a relatively high coefficient of variation (0.20385), indicating moderate dispersion, while P_{S2} exhibited a narrower range (80–160 kPa) but a slightly larger relative variability (0.27986). T_{EVP} and T_{PEM} showed approximately identical variance and

Table 1
Descriptive statistics of operational inputs and performance outputs used for modeling and optimization of the integrated geothermal-based tri-generation system.

Statistical Parameters	Input						Target		
	P _{S1} (kPa)	P _{S2} (kPa)	T _{EVp} (°C)	P _{LNG} (kPa)	α	T _{PEM} (°C)	EE (%)	PSC (\$/GJ)	MFR _{H2} (kg/h)
Minimum	600	80	162	2500	0.25	70	40.71	19.68	1.37
Maximum	1000	160	182	3500	0.75	90	44.42	21.84	11.37
Mean	810	118	170.5	3100	0.525	81.5	43.156	20.425	6.3351
Median	800	120	172	3000	0.5	80	43.41	20.265	6.53
Variance	27,263.2	1090.53	66.0526	173,684.2	0.0388	66.053	1.1156	0.3890	8.9940
Standard deviation	165.116	33.0231	8.12728	416.7544	0.1970	8.1273	1.0562	0.6237	2.9990
Coefficient of variation	0.20385	0.27986	0.04767	0.13444	0.3753	0.0997	0.0245	0.0305	0.4734
Skew	-0.098	0.098	0.296	-0.412	-0.186	-0.296	-1.25	1.317	0.537
Kurtosis	-1.518	-1.518	-1.399	-1.434	-1.308	-1.399	1.019	0.951	-0.302
Kolmogorov-Smirnov stat	0.225	0.225	0.252	0.281	0.223	0.252	0.196	0.248	0.273

standard deviation values, both reflecting controlled operating ranges (162–182 °C and 70–90 °C, respectively) and low variability (CV ≈ 0.048 and 0.0997). P_{LNG} displayed the highest variance (173,684.2) due to its wide operational span (2500–3500 kPa), yet its coefficient of variation (0.13444) remained moderate, suggesting consistent operation relative to its scale. The dimensionless parameter α, representing the work fraction allocated to electrolysis, showed the largest relative

dispersion among inputs (CV = 0.3753), emphasizing its strong variability and high potential influence on hydrogen production. Regarding the distribution characteristics, the skewness values revealed that P_{S1} and P_{S2} exhibited almost symmetric behavior (−0.098 and 0.098), T_{EVp} showed moderate right-skewed trend (0.296), P_{LNG} and T_{PEM} presented a mild left-skewed patterns (−0.412 and −0.296), and α demonstrated slight left skewness (−0.186), indicating that all input variables were

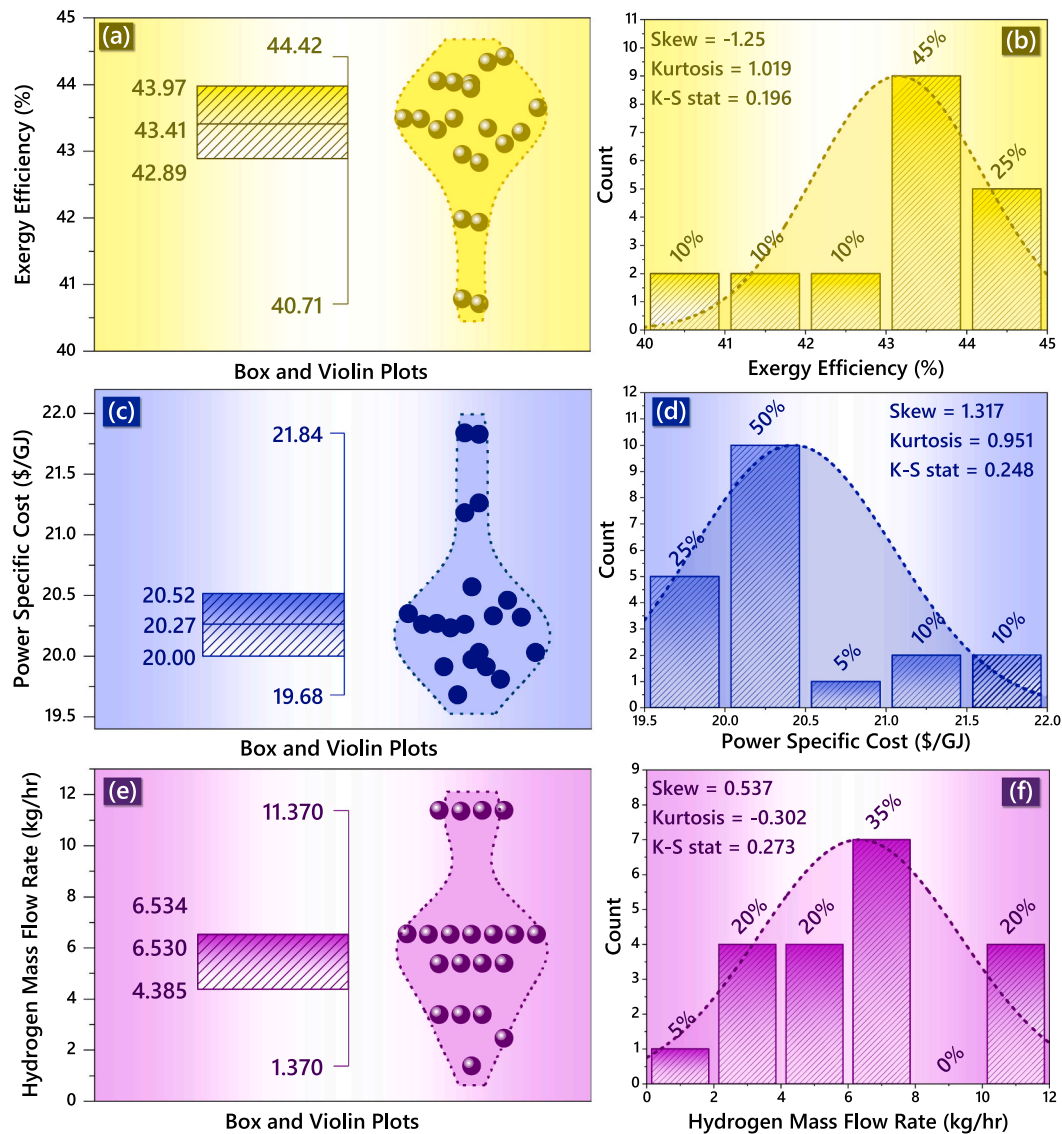


Fig. 3. Data distribution of the three performance targets of the integrated tri-generation system, presented through combined box–violin plots and frequency histograms: (a–b) EE, (c–d) PSC, and (e–f) MFR_{H₂}.

relatively close to symmetric distributions with minor deviations in either direction. All input variables exhibited negative kurtosis, pointing to flatter, less peaked distributions compared to a normal curve, which aligns with Kolmogorov-Smirnov statistics confirming mild deviations from normality.

For the targets, EE varied between 40.71% and 44.42%, exhibiting low relative dispersion ($CV = 0.0245$) and strong left skew (-1.25), suggesting concentration toward higher efficiencies. PSC showed a narrow variation (19.68 – 21.84 \$/GJ) with low variability ($CV = 0.0305$) but strong positive skew (1.317), indicating a tendency toward lower cost values with occasional high-cost cases. Hydrogen production rate presented the highest spread (1.37 – 11.37 kg/h) and substantial variability ($CV = 0.4734$), reflecting strong sensitivity to operational changes, with moderate right skew (0.537) and slightly flatter distribution (kurtosis = -0.302).

These descriptive statistics provide a quantitative foundation for evaluating the behavior of the target outputs by indicating how efficiently, economically, and consistently the system responded under different operating conditions. The observed spreads and distribution metrics enable the identification of outputs that are more sensitive to operational variations, such as hydrogen production, compared to more stable indicators like exergy efficiency. By clarifying which variables fluctuate widely and which remain consistent, these statistics support the selection of appropriate modeling strategies and guide optimization toward regions where improvements are both feasible and impactful. Consequently, the reliability of predictive models increases, as they are built on outputs whose variability and distribution are clearly understood in advance.

Fig. 3 illustrates the data distribution of the three performance targets, supporting the statistical trends presented in Table 1. The combined box and violin plots highlight the spread, central tendency, and density of data points, while the frequency histograms reveal how frequently values occur across each range. For exergy efficiency (Fig. 3 (a–b)), the data are tightly concentrated between approximately 42.8% and 44.4%, matching the narrow variability implied by the descriptive statistics. This cluster creates a left-skewed distribution, confirming that most values lean toward the higher end of the efficiency range. The histogram indicates that 45% of observations fall between 43 and 44%, making it the most dominant interval, while about 25% and 10% of values lie in adjacent ranges. The density pattern in the violin plot further verifies the strong probability of values accumulating near the median, supporting the low dispersion reflected in Table 1.

In contrast, the PSC distribution (Fig. 3(c–d)) exhibits greater spread, with values scattered between roughly \$19.7/GJ and \$21.8/GJ. The histogram reveals a right-skewed tendency, indicating that lower costs are more frequent than the higher values skewing the tail. Approximately 50% of the data lie between \$20–\$20.5/GJ, demonstrating that most system configurations remain cost-effective under this interval. The violin shape confirms a high-density region within this band, consistent with the moderate variability identified statistically.

Hydrogen production (Fig. 3(e–f)) shows the widest dispersion among targets. Although the box plot ranges from about 1.37 to 11.37 kg/h, the histogram indicates 35% of observations clustered around 6–8 kg/h, making it the most frequent operating window. The mildly right-skewed distribution implies that larger production rates occur less frequently but significantly influence system optimization potential. The violin plot supports this distribution, displaying dense clustering near the median and sparser tails.

2.3. Correlation analysis

The Pearson correlation coefficient (PCC) is a statistical index that measures the strength and direction of a linear association between two quantitative variables [37]. In engineering studies, including the present investigation, PCC is essential for identifying how one parameter linearly influences another, thereby guiding design choices before

optimization and modeling. Its mathematical definition is expressed as:

$$PCC = \frac{Cov(X, Y)}{\sigma_X \sigma_Y} \quad (1)$$

Here, $Cov(X, Y)$ represents the covariance between the variables X and Y , while σ_X and σ_Y denote their corresponding standard deviations. This metric provides an initial basis for identifying influential relationships among operational parameters and system responses.

Fig. 4 presents the PCC values among the inputs and targets of a geothermal-driven tri-generation system, revealing key linear dependencies. For input–input interactions, T_{EVP} shows a moderate positive correlation with P_{S2} (30%), while it exhibits a notable negative relationship with T_{PEM} (–44%), emphasizing thermal coupling between the evaporator and membrane unit. Regarding input–output links, T_{EVP} demonstrates a strong negative PCC with PSC (–48%) and a very high negative association with MFR_{H_2} (–73%), while maintaining a moderately positive correlation with EE (40%). P_{LNG} also shows meaningful linear effects, displaying substantial positive correlation with EE (53%) and strong negative correlation with PSC (–47%). The parameter α significantly influences all target responses, indicated by a strong positive correlation with both PSC (65%) and MFR_{H_2} (45%), whereas its correlation with EE is highly negative (–68%), showing conflicting impacts between economic and energetic performance. Among outputs, EE has an almost perfect negative PCC with PSC (–99%) and strong negative association with MFR_{H_2} (–81%), while PSC and MFR_{H_2} display a very high positive correlation (88%), suggesting that higher hydrogen production increases cost. Conversely, some coefficients remain weak or negligible (for example, P_{S1} with P_{S2} and EE), but these should not be disregarded, since a lack of linear correlation does not eliminate the possibility of strong nonlinear dependencies. Therefore, advanced machine learning techniques are highly advantageous for uncovering such hidden nonlinear interactions and improving system optimization.

3. Machine learning

After establishing the descriptive and correlation analyses of the system inputs and outputs, the machine learning approach was employed for regression modeling of the performance objectives using the available dataset from the geothermal-based tri-generation system. Initially, the study presents the selected machine learning algorithm and the meta-heuristic optimization techniques employed to tune its hyperparameters, thereby ensuring accurate and reliable predictions. The model development process involved data preprocessing, training, and testing, with performance assessed through statistical metrics. Finally, the models were systematically compared, and the most effective algorithm for each objective was selected for use in the subsequent optimization process.

3.1. Multilayer perceptron neural networks

The multilayer perceptron neural network is a widely used feedforward artificial neural network designed for both classification and regression tasks, characterized by its layered structure comprising an input layer, one or more hidden layers, and an output layer [38]. Each neuron in a given layer applies a nonlinear transfer function to a weighted sum of its inputs, enabling the network to capture complex, nonlinear relationships between variables. MLPNNs are particularly effective in regression modeling due to their capacity to approximate continuous functions with high precision and their robustness in handling multivariate datasets with interdependent features [39]. Key factors influencing MLPNN performance include the number of hidden layers and neurons, choice of transfer functions, learning rate, momentum, and training algorithms [40]. ANN parameters are categorized into training-related parameters, such as the learning rate, training algorithm, and number of epochs, and structural parameters, including network depth and width, as reported in previous studies [41,42]. This

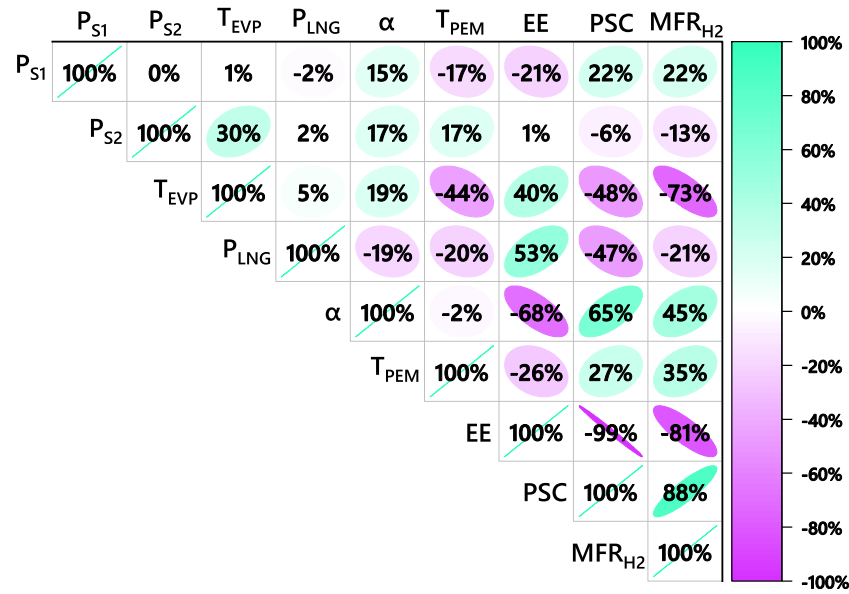


Fig. 4. Pearson correlation coefficients among the input parameters and target outputs of the geothermal-based tri-generation system.

combination of adaptability and computational efficiency makes MLPNNs a preferred tool for predictive modeling in engineering systems.

3.2. Particle swarm optimization (PSO)

PSO is a population-based metaheuristic algorithm commonly applied for tuning MLPNN hyperparameters to enhance prediction accuracy and convergence stability [43]. Inspired by the collective behavior of flocks of birds or schools of fish, PSO represents each solution as a particle that adapts its trajectory based on its own experience and the best positions identified by the swarm. The process starts with a randomly generated set of particles, whose fitness values are assessed at each iteration. Particle positions and velocities are subsequently updated by combining information from the individual's best-known position (pBest) and the swarm's overall best position (gBest), steering the population toward optimal solutions. This update cycle repeats iteratively until the algorithm satisfies a predefined stopping condition or reaches convergence. PSO is valued for its conceptual simplicity, ability to operate without gradient information, and effective exploration of complex, nonlinear, and high-dimensional search spaces, often surpassing conventional evolutionary approaches in engineering optimization and machine learning tasks, such as tuning the parameters of machine learning models, as reported in previous studies [44,45].

3.3. Marine predators algorithm (MPA)

MPA is a population-based optimization technique inspired by the foraging strategies of marine animals [46]. In this approach, candidate solutions are represented as interacting predators and prey, with movement strategies designed to maintain a balance between global search and local refinement. The algorithm initiates with a randomly generated population of prey, and the best-performing individual is designated as the elite predator [47]. During the initial stages, the search emphasizes exploration, with prey following Brownian motion patterns to survey the solution space while predators track and influence the swarm. As iterations progress, the algorithm alternates between wide-ranging exploration and intensive local exploitation through Lévy flight-based movements, promoting convergence toward high-quality regions [48]. Adaptive step sizes and mechanisms simulating environmental features, such as fish aggregating devices (FADs), help preserve population diversity and prevent premature stagnation. The elite matrix

is continuously updated with superior solutions, enabling MPA to effectively address complex, multimodal, and high-dimensional optimization challenges. This algorithm has demonstrated strong performance in a limited number of studies focused on optimizing the parameters of machine learning models [49].

3.4. White shark optimizer (WSO)

After drawing inspiration from the predatory strategies of white sharks, the WSO is formulated as a population-based metaheuristic for solving complex optimization problems [50]. WSO mimics the remarkable hunting abilities of great white sharks, particularly their exceptional senses of hearing, sight, and smell, as well as their undulating swimming patterns and group behavior. These biological traits are mathematically abstracted to guide artificial search agents toward optimal solutions in a given search space. The algorithm balances exploration and exploitation by integrating three primary behaviors: undulating motion toward detected prey, random searching in unexplored regions, and coordinated movement based on the best-performing agents, reflecting fish school dynamics.

Initially, the WSO randomly generates a population of candidate solutions within the defined search domain. Each agent, representing a white shark, is assigned a velocity vector and a position in the multi-dimensional space. The fitness of each solution is evaluated using the problem-specific objective function. Iteratively, WSO updates the velocity and position of each agent by considering the influence of the global best, personal best, and stochastic components, combined with adaptive parameters controlling exploration and exploitation. As iterations progress, the strength of the agent's sensing capability, modeled as the parameter mv , increases, enabling the algorithm to shift from global exploration to local refinement around promising regions. The elite solutions are preserved, and collective behaviors ensure agents exploit the vicinity of high-quality solutions while avoiding premature convergence. The step-by-step procedure of the WSO optimization process is [50,51]:

1. Initialize the problem and algorithm parameters, including population size and search space dimensions.
2. Randomly generate the initial positions of the search agents.
3. Initialize the velocities of all agents.
4. Evaluate the fitness of each initial solution.
5. Repeat until the maximum number of iterations is reached:

- a. Update adaptive parameters that control velocity, sensing, and search direction.
 - b. Adjust agent velocities based on global and personal best positions.
 - c. Update agent positions using either random local search or undulating motion toward prey.
 - d. Move agents toward the best-performing agent to simulate collective behavior.
 - e. Handle any boundary violations and evaluate the updated positions.
 - f. Update elite and global best solutions.
6. Increment the iteration counter.
 7. Return the best solution obtained.

WSO effectively combines biologically inspired strategies and adaptive parameter control, offering robust convergence, enhanced exploration of complex landscapes, and reliable identification of global optima in multimodal, nonlinear optimization tasks.

3.5. Models development and assessment

The MLPNN model developed in this work was designed with careful attention to both its architecture and training protocol. Architecturally, due to the limited dataset size, the neural network design was intentionally kept minimalistic to optimize robustness and prevent overfitting. A unidirectional, layered architecture with a single hidden layer was employed, with the maximum neuron count set at 100. This constrained configuration provides sufficient capacity to capture complex nonlinear dynamics while controlling the total number of parameters. Such an approach strikes an optimal balance between the network's expressive power and the statistical limitations inherent in the small sample set.

Various transfer functions were tested for both hidden and output layers, including linear, sigmoid, radial basis, saturating linear, and competitive functions, allowing the network to represent highly nonlinear thermal dynamics with accuracy. Training focused on optimizing critical parameters, such as learning rate and the training strategy. To explore these hyperparameters comprehensively, a variety of training approaches were employed, encompassing conventional gradient-based methods, conjugate gradient variants, resilient back-propagation, one-step secant, quasi-Newton methods, Levenberg–Marquardt, and Bayesian regularization techniques.

The surrogate modeling approach was developed using a dataset consisting of 20 observations generated from detailed thermodynamic simulations performed in the EES environment. Because of the inherently small number of samples, the modeling procedure was designed to operate effectively under data-scarce conditions. To evaluate predictive capability in an unbiased manner, the dataset was divided into two independent subsets: 80% of the samples were allocated to the training stage, while the remaining 20% were reserved exclusively for external performance assessment. Despite the modest dataset size, this partitioning allows the generalization ability of the model to be examined on unseen cases. Furthermore, during the training phase, a validation portion corresponding to 30% of the training data was incorporated into the learning workflow. This validation set was used within an early stopping mechanism to continuously track prediction errors, restrain excessive model fitting, and control network complexity, ultimately promoting stable training behavior and reliable convergence of the MLPNN model.

For reproducibility, the initialization of the neural network parameters was controlled using a fixed random seed implemented in MATLAB R2024a through the command `rng('default')`. For each tuning technique, a maximum of 2000 evaluations of the objective function was performed, and the optimization process was independently executed five times to capture the effects of algorithmic randomness.

Model performance was assessed using five widely adopted statisti-

cal metrics. The mean squared error (MSE) measured the squared differences between predicted outputs and reference data, whereas the mean absolute percentage error (MAPE) represented prediction deviations as a percentage of the true values, enabling comparisons independent of data magnitude. The correlation coefficient (R) was employed to evaluate the strength of the linear relationship between model outputs and reference data, while the coefficient of determination (R^2) quantified the fraction of variability in the observed responses explained by the predictive model. Willmott's agreement index (I_A) was used to quantify the level of consistency between simulated results and experimental data, with values spanning from zero, indicating no correspondence, to unity, representing complete agreement. These criteria are well known for assessing machine learning models and have been widely used in many previous studies [52–55]:

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_{i,Act} - Y_{i,Pred})^2 \quad (2)$$

$$MAPE(\%) = \frac{1}{n} \sum_{i=1}^n \left| \frac{Y_{i,Pred} - Y_{i,Act}}{Y_{i,Act}} \right| \times 100 \quad (3)$$

$$R = \frac{\sum_{i=1}^n (Y_{i,Act} - \bar{Y}_{i,Act})(Y_{i,Pred} - \bar{Y}_{i,Pred})}{\sqrt{\sum_{i=1}^n (Y_{i,Act} - \bar{Y}_{i,Act})^2 \sum_{i=1}^n (Y_{i,Pred} - \bar{Y}_{i,Pred})^2}} \quad (4)$$

$$R^2 = 1 - \sum_{i=1}^n \frac{(Y_{i,Pred} - Y_{i,Act})^2}{Y_{i,Act}^2} \quad (5)$$

$$I_A = 1 - \frac{\sum_{i=1}^n (Y_{i,Act} - Y_{i,Pred})^2}{\sum_{i=1}^n (|Y_{i,Pred} - \bar{Y}_{i,Act}| + |Y_{i,Act} - \bar{Y}_{i,Act}|)^2} \quad (6)$$

Collectively, MSE and MAPE quantified prediction errors, while R, R^2 , and I_A reflected model fidelity and robustness. High values of R, R^2 , and I_A confirm that the MLPNN, when optimized via PSO, MPA, and WSO, can effectively capture the complex performance dynamics of the geothermal-based tri-generation system under varying operating conditions. Here, $Y_{i,Act}$ and $Y_{i,Pred}$ represent observed and predicted values, respectively, and n is the total number of samples.

3.6. Models evaluation

Table 2 presents a comprehensive overview of the performance metrics of the developed MLPNN models optimized via PSO, MPA, and WSO for predicting the key performance indicators. Focusing on the testing dataset, which provides the most relevant insight into model generalization, the PSO-optimized MLPNN exhibited exceptional predictive capability for EE, achieving an MSE of 3.79×10^{-4} and a MAPE of 0.027%, alongside near-perfect correlation coefficients ($R = 0.99996$) and coefficient of determination ($R^2 = 0.99993$), with a corresponding I_A of 0.99995, indicating outstanding accuracy and robustness. In comparison, MPA achieved slightly higher errors ($MSE = 1.18 \times 10^{-3}$, $MAPE = 0.054\%$) and marginally lower correlation and I_A values, while WSO, although delivering acceptable performance, demonstrated the largest deviation among the three optimizers for EE ($MSE = 1.67 \times 10^{-2}$, $MAPE = 0.206\%$), suggesting a comparatively less precise mapping of the EE output. For PSC, the MPA-based optimization outperformed both PSO and WSO on the test set, with an exceptionally low MSE of 2.04×10^{-5} and a MAPE of 0.015%, accompanied by R and R^2 values of 0.99993 and 0.99986, respectively, reflecting highly reliable predictions. PSO, while robust ($MSE = 2.50 \times 10^{-3}$, $MAPE = 0.149\%$), was less accurate, and WSO again showed comparatively higher error metrics ($MSE = 5.48 \times 10^{-3}$, $MAPE = 0.254\%$), though still maintaining

Table 2

Statistical performance of MLPNN models optimized with PSO, MPA, and WSO for training and testing datasets across all objectives.

Objective	Optimizer	Dataset	MSE	MAPE (%)	R	R ²	I _A
EE	PSO	Testing	3.79E-04	0.027	0.99996	0.99993	0.99995
		Training	3.41E-05	0.005	0.99999	0.99997	0.99999
	MPA	Testing	1.18E-03	0.054	0.99982	0.99964	0.99984
		Training	3.01E-04	0.014	0.99990	0.99981	0.99992
	WSO	Testing	1.67E-02	0.206	0.99751	0.99502	0.99768
		Training	3.37E-04	0.017	0.99981	0.99963	0.99991
PSC	PSO	Testing	2.50E-03	0.149	0.99973	0.99946	0.99894
		Training	1.08E-03	0.074	0.99914	0.99829	0.99912
	MPA	Testing	2.04E-05	0.015	0.99993	0.99986	0.99992
		Training	2.72E-05	0.012	0.99996	0.99992	0.99996
	WSO	Testing	5.48E-03	0.254	0.99827	0.99654	0.99769
		Training	1.01E-03	0.076	0.99888	0.99776	0.99918
MFR _{H2}	PSO	Testing	5.59E-03	0.775	0.99980	0.99960	0.99984
		Training	4.14E-04	0.374	0.99984	0.99968	0.99998
	MPA	Testing	8.06E-02	2.835	0.99836	0.99673	0.99747
		Training	1.85E-01	9.281	0.99095	0.98198	0.99465
	WSO	Testing	8.74E-04	0.262	0.99992	0.99983	0.99997
		Training	3.61E-04	0.109	0.99998	0.99990	0.99999

strong correlation and agreement indices. Regarding hydrogen production rate, MFR_{H2}, the WSO-optimized model demonstrated superior predictive performance on the test dataset, achieving MSE = 8.74×10^{-4} , MAPE = 0.262%, R = 0.99992, R² = 0.99983, and I_A = 0.99997, clearly outperforming both PSO and MPA. The PSO model, while highly competent (MSE = 5.59×10^{-3} , MAPE = 0.775%), was moderately less precise, and the MPA model exhibited the largest deviations (MSE = 8.06×10^{-2} , MAPE = 2.835%), suggesting limited generalization in predicting hydrogen production. Across all objectives, it is evident that each metaheuristic algorithm exhibits varying strengths depending on the specific output, highlighting the importance of selecting optimizer strategies tailored to each system target, while noting that the training performance of all models was consistently excellent, confirming reliable learning during model development. The spider plots presented in Fig. 5 visually summarize these findings, clearly illustrating that the best-optimized models achieve minimal error and maximal correlation across the three objectives on the testing dataset, providing a rapid comparative visualization of relative performance. The results highlight the robustness, generalization, and reliability of the proposed modeling framework, ensuring that the optimized MLPNN models can reliably predict system behavior under varied operating conditions, thereby supporting optimal design and operational decision-making for geothermal-based tri-generation systems while efficiently balancing exergy efficiency, cost, and hydrogen production objectives.

The exceptionally high predictive accuracy reported in Table 2 is a direct consequence of the deterministic nature of the dataset and the safeguards implemented throughout the modeling process, rather than an indication of overfitting. The data were generated from high-fidelity thermodynamic simulations in EES, where all outputs strictly follow mass, energy, and exergy conservation principles. As such, the input–output relationships are noise-free and fully governed by deterministic physical laws, unlike experimental datasets that embed stochastic variability. In this smooth and highly structured design space, an MLPNN can accurately learn the underlying functional manifolds, naturally yielding very high R² and low MAPE values. Moreover, the simulation points were generated using a structured design-of-experiments methodology to ensure full coverage of the geothermal–LNG operating domain. This space-filling sampling strategy prevents the surrogate models from extrapolating into sparsely populated regions and enhances their ability to generalize within the defined parameter bounds. To further avoid overfitting, strict methodological controls were applied. The MLPNN architecture was intentionally constrained to a single hidden layer with no more than 100 neurons, and model training employed an 80/20 data split with 30% of the training portion used for real-time validation and early stopping. Comprehensive evaluation using multiple error metrics revealed no signs of overfitting.

Additionally, the use of metaheuristic optimization algorithms (PSO, MPA, WSO) ensured high accuracy by globally tuning structural/training MLPNN parameters. Collectively, these factors justify the high accuracy achieved while confirming the robustness and generalization capability of the developed models.

Fig. 6 illustrates the performance of the optimized MLPNN models through three complementary sets of diagnostics (regression plots, combined violin/box distributions, and absolute relative deviation (ARD) polar charts) providing a comprehensive picture of model accuracy for both training and testing samples. In the regression graphs, the predicted values of EE, PSC, and MFR_{H2} are plotted against their corresponding actual values, with the 45° line (Y = X) serving as the reference for perfect predictions. For all three models (PSO-ANN for EE, MPA-ANN for PSC, and WSO-ANN for MFR_{H2}), both training and test points cluster extremely close to this line, and the minimal lateral deviations indicate negligible prediction bias. The testing data points show nearly the same proximity to the Y = X line as the training points, confirming excellent generalization capability with no observable overfitting. This is further supported by the extremely high correlation coefficients (all above 0.9999), which verify that the models reproduce the physical relationships between process inputs and performance indicators with very high fidelity.

The mixture plots of violin and box distributions provide an additional statistical comparison between the actual and predicted datasets for each target variable. The violin shapes of actual and predicted values nearly overlap in all cases, demonstrating that the MLPNN models accurately capture not only the central tendencies but also the full distributional behavior of the outputs. The predicted distributions for EE, PSC, and MFR_{H2} follow the same spread, symmetry, and density patterns observed in the actual data, indicating that the global trends and variability of the system are well represented. The box plots embedded within each violin further corroborate this: the medians of actual and predicted data are virtually identical, the interquartile ranges show strong alignment, and the whiskers span nearly the same limits. Only a few sparse points appear near the edges, and even these minor deviations fall well within acceptable modeling uncertainty, highlighting the stability and robustness of the trained networks.

Before interpreting the ARD polar charts, the ARD itself is defined as a normalized error metric given by:

$$\text{ARD} = \left| \frac{y_{\text{actual}} - y_{\text{predicted}}}{y_{\text{actual}}} \right| \times 100\%. \quad (7)$$

This metric quantifies the relative percentage deviation of each prediction from its true value and is particularly useful for evaluating model precision across variables of different scales. The ARD polar charts summarize the distribution of ARD values for all training and test

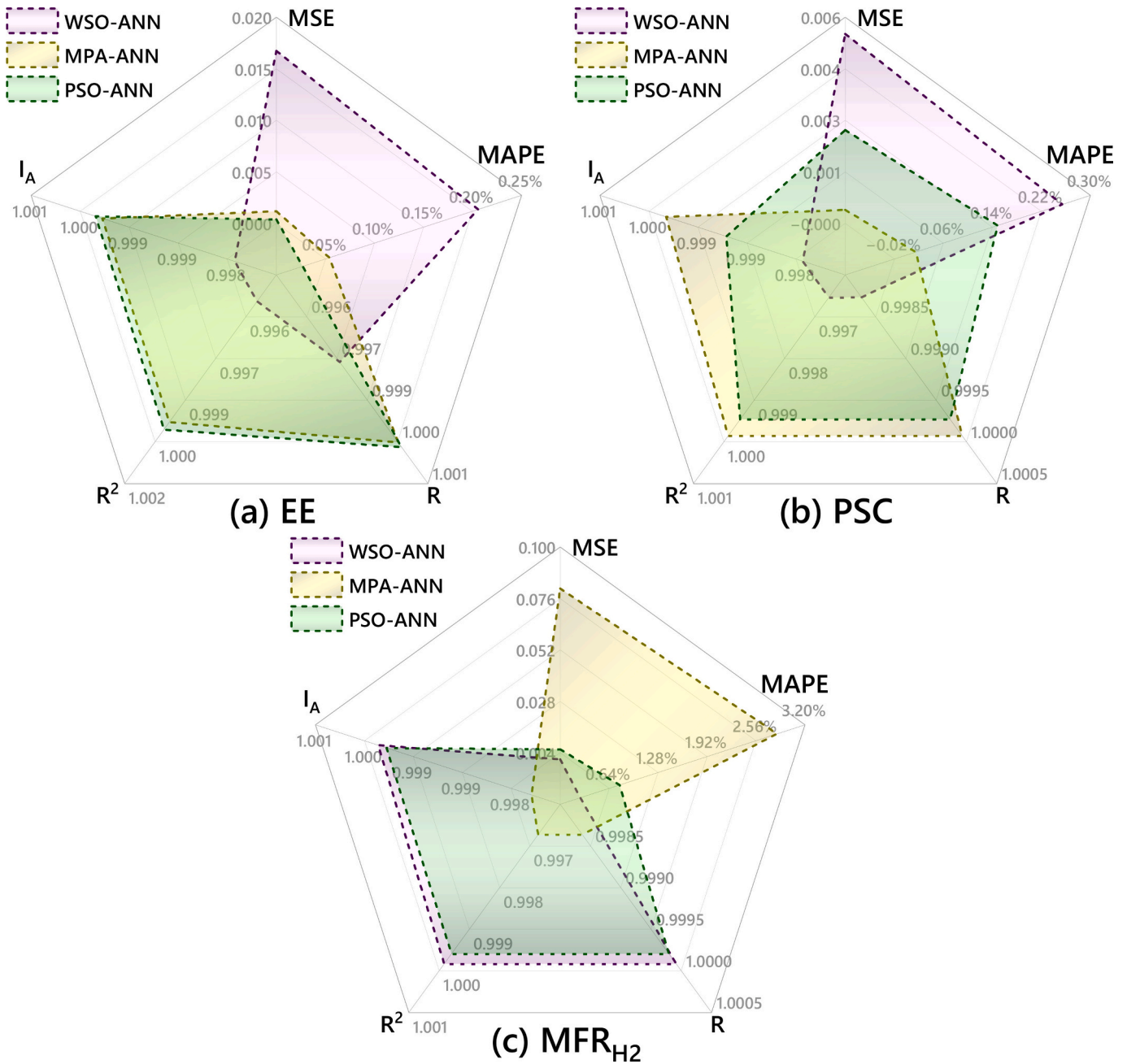


Fig. 5. Spider plots comparing the performance of the best MLPNN models on the testing dataset.

samples across the three target variables. For EE, PSC, and MFR_{H_2} , the ARD magnitudes are overwhelmingly concentrated in extremely small ranges, with most deviations falling far below 1%. Based on the charts, only a few scattered points exhibit deviations slightly above 1%, while the vast majority of the data remain well below the 1% threshold, confirming the high precision and stability of the developed models.

These results confirm that the optimized MLPNN structures can model the complex thermo-economic-hydrogen-production behavior of the geothermal-based tri-generation system with exceptionally high accuracy. This robust predictive capability provides a strong and reliable foundation for subsequent multi-objective optimization, enabling confident exploration of trade-offs among exergy efficiency, power-specific cost, and hydrogen production rate.

Fig. 7 presents a comprehensive sensitivity analysis illustrating how variations in key operational inputs influence the system's three primary performance indicators, as predicted by the top-performing optimized

MLPNN models. In each contour map, two variables are varied while all remaining inputs are held constant to isolate their individual and combined effects. For EE, the interactions shown in Figs. 7(a–c) demonstrate that the efficiency strongly depends on the pressure levels within the separators and the thermal conditions of the cycle. In Fig. 7(a), the influence of separator pressures on EE demonstrates contrasting behavior: increasing P_{S1} leads to a gradual decline in EE, whereas increasing P_{S2} enhances EE significantly, indicating that the second separation stage contributes more strongly to cycle thermodynamics. Fig. 7(b) shows that both T_{EVP} and P_{LNG} exert a positive and monotonic influence on EE, where higher evaporator temperatures strengthen thermal energy availability and higher LNG pump pressures boost cold-energy recovery, jointly increasing the system's overall exergy performance. In Fig. 7(c), EE decreases noticeably with increasing α , especially beyond mid-range values, because diverting more ORC power toward electrolysis reduces the net exergy efficiency; meanwhile, variations in T_{PEM} show only a

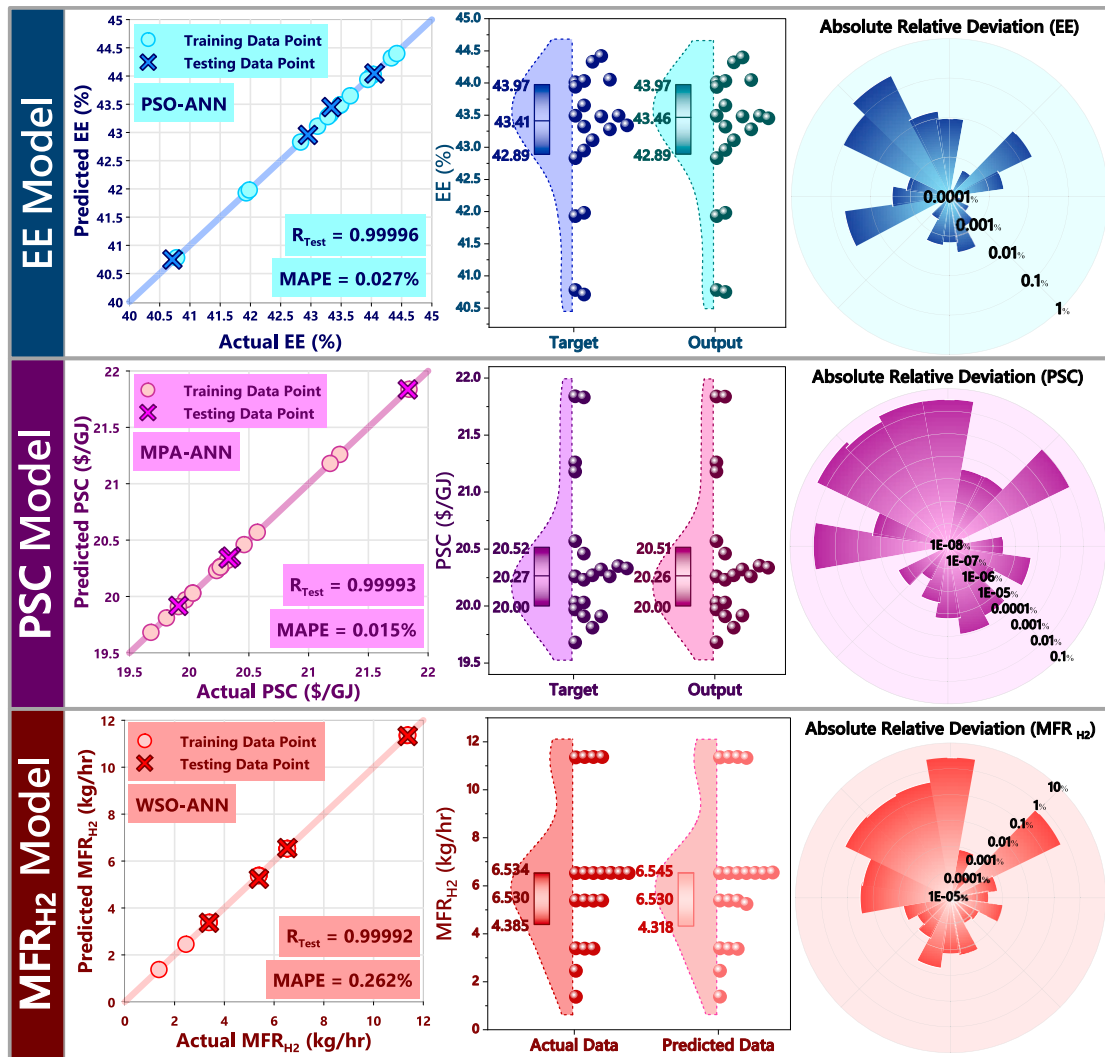


Fig. 6. Regression fit, violin-box distribution, and ARD deviation plots for the optimized MLPNN models predicting exergy efficiency, power-specific cost, and hydrogen production.

minor positive influence on EE, indicating that membrane temperature has a secondary but still beneficial effect on electrochemical performance. The behavior of PSC in Fig. 7(d) is more complex: medium-to-high values of P_{S2} sharply reduce PSC, signaling improved cost effectiveness, yet in the low-pressure region the value of P_{S1} becomes decisive: when P_{S1} lies between 700 and 800 kPa, PSC rises steeply, highlighting a sensitive zone where small pressure changes lead to substantial cost fluctuations. In Fig. 7(e), both T_{EVP} and P_{LNG} consistently decrease PSC, although the influence of T_{EVP} is noticeably stronger, indicating that improvements in evaporator temperature more effectively enhance thermal matching and reduce unit cost compared to the more modest contribution of increased LNG pump pressure. Fig. 7(f) shows that increases in α , particularly beyond 0.6, cause PSC to rise dramatically due to the higher energetic burden imposed by electrolysis, whereas increasing T_{PEM} exerts little influence on PSC, indicating that membrane temperature is thermodynamically important but economically marginal. For the hydrogen production rate, Fig. 7(g) reveals that both P_{S1} and P_{S2} positively affect MFR_{H2} , with higher pressures improving flow transport and electrolyzer feed conditions. In contrast, Fig. 7(h) shows a strong negative dependence on T_{EVP} , where rising evaporator temperature substantially reduces MFR_{H2} because higher thermal conditions shift the ORC toward power production rather than hydrogen generation; meanwhile, P_{LNG} has only a minor effect, reflecting the secondary role of LNG pumping on electrolysis performance.

Finally, Fig. 7(i) shows that α is the dominant driver of hydrogen production. MFR_{H2} increases sharply beyond $\alpha = 0.5$ as more ORC power is allocated to the electrolyzer. T_{PEM} increases MFR_{H2} up to approximately $\alpha = 0.5$ due to enhanced membrane conductivity, but beyond this range its influence becomes negligible as α becomes the overriding control parameter. Collectively, these sensitivity patterns highlight the nonlinear and highly coupled nature of the geothermal tri-generation system and emphasize the importance of precise multi-variable coordination for achieving optimal efficiency, cost, and hydrogen productivity. Given these complex trade-offs, the system clearly requires a robust multi-objective optimization framework to simultaneously balance thermodynamic performance, economic feasibility, and hydrogen output, ensuring an optimal and practical operating strategy.

4. Multi-objective optimization

Enhancing the performance of the proposed geothermal-LNG-hydrogen tri-generation system requires a careful balance among conflicting objectives, including maximizing exergy efficiency, minimizing power-specific cost, and increasing hydrogen production rate. These targets are inherently correlated; for example, improving efficiency may require additional energy recovery units that increase production cost, while maximizing hydrogen yield might alter operating conditions in a way that reduces total exergy performance. Because

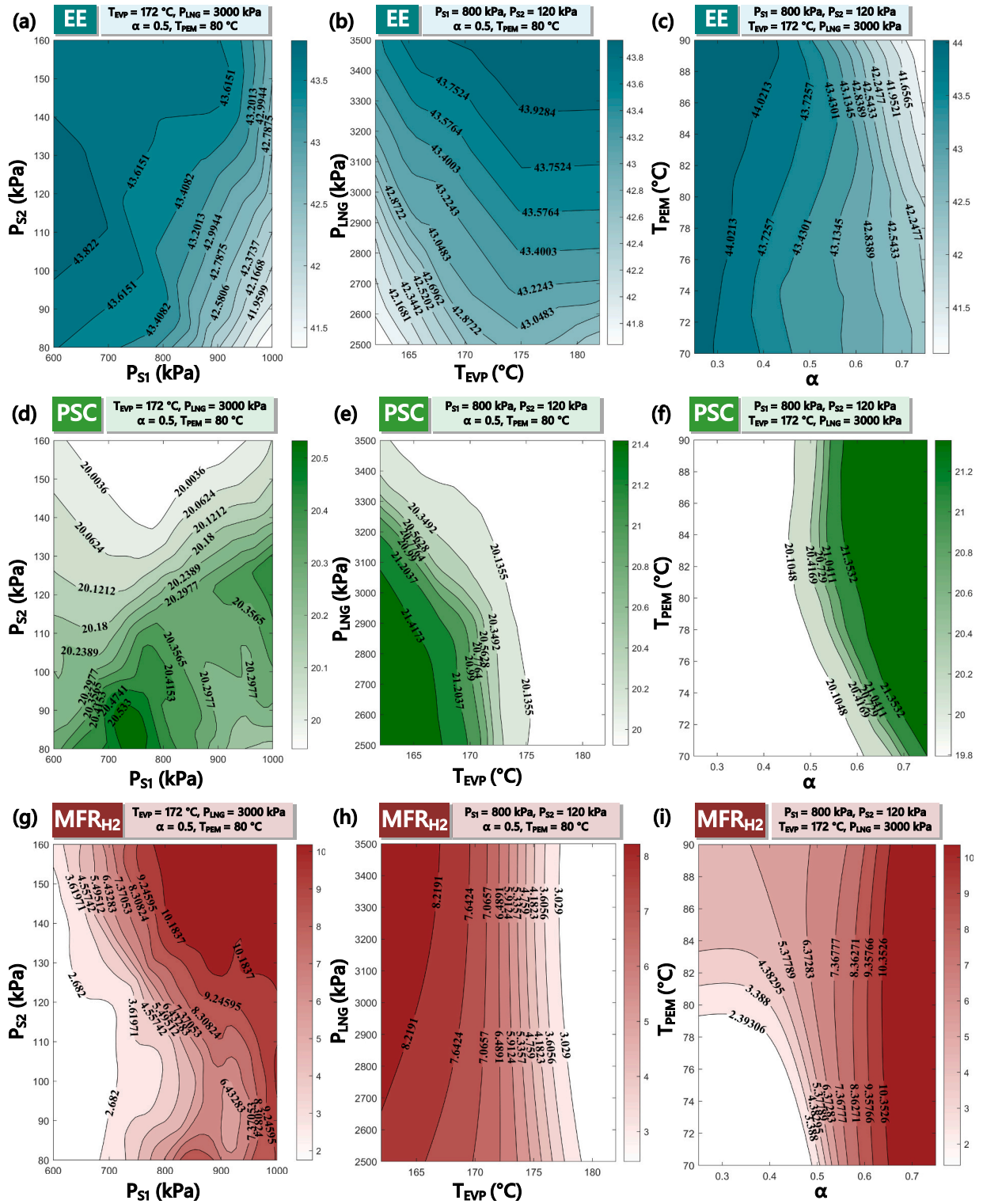


Fig. 7. Sensitivity contours showing the effects of key input variables on exergy efficiency, power-specific cost, and hydrogen production rate.

improvements in one objective can detrimentally influence the others, it is impractical to pursue a single universal optimum. Instead, achieving a realistic and reliable design solution demands a comprehensive multi-objective optimization strategy capable of simultaneously considering all trade-offs. To address this need, the present study utilizes a multi-objective material generation algorithm combined with the optimized MLPNN predictive models to explore the optimal operational and thermodynamic settings of the tri-generation cycle. MOMGA generates a set

of Pareto-optimal solutions that reflect competing configurations, each representing a unique compromise among the three performance measures. This approach enables thorough exploration of the decision space and facilitates selection of operational designs that achieve the most desirable balance among energy performance, economic feasibility, and hydrogen synthesis. The multi-objective optimization problem is formulated as follows:

$$\begin{aligned}
& \begin{cases} \text{Maximize } EE = f(P_{S1}, P_{S2}, T_{EVP}, P_{LNG}, \alpha, T_{PEM}) \\ \text{Minimize } PSC = g(P_{S1}, P_{S2}, T_{EVP}, P_{LNG}, \alpha, T_{PEM}) \\ \text{Maximize } MFR_{H2} = h(P_{S1}, P_{S2}, T_{EVP}, P_{LNG}, \alpha, T_{PEM}) \end{cases} \\
& \text{Subject to } \begin{cases} 600 \leq P_{S1} \leq 1000 \text{ (kPa)} \\ 80 \leq P_{S2} \leq 160 \text{ (kPa)} \\ 162 \leq T_{EVP} \leq 182 \text{ (}^\circ\text{C)} \\ 2500 \leq P_{LNG} \leq 3500 \text{ (kPa)} \\ 0.25 \leq \alpha \leq 0.75 \\ 70 \leq T_{PEM} \leq 90 \text{ (}^\circ\text{C)} \end{cases} \quad (8)
\end{aligned}$$

Here, f , g , and h represent the MLPNN-based surrogate models corresponding to exergy efficiency, power-specific cost, and hydrogen mass flow rate, respectively.

4.1. Multi-objective material generation algorithm (MOMGA)

The material generation algorithm (MGA) is a recently developed physics-inspired metaheuristic that models optimization processes based on the mechanisms through which natural materials are created and transformed [56]. Instead of imitating biological evolution, MGA relies on fundamental concepts from chemistry and material science. The algorithm views each candidate solution in the decision space as a “material” composed of hypothetical periodic elements. Analogous to atoms combining to form complex matter, the decision variables in MGA behave like constituents of chemical structures, interacting through bonding, excitation, sharing, and exchange of properties. During initialization, each candidate solution is assigned random pseudo-atomic values within predefined ranges, representing different possible material compositions. Mathematically, the first population is expressed as:

$$\text{Mat} = [PTE_i^j]_{n \times d} \quad (9)$$

$$PTE_i^j(0) = PTE_{i,\min}^j + \text{Unif}(0, 1) \left(PTE_{i,\max}^j - PTE_{i,\min}^j \right) \quad (10)$$

where n represents the number of “materials” (solutions), d is the dimensionality of the optimization problem, and the term PTE_i^j denotes the quantity of the j -th variable within the i -th solution.

Once initialized, MGA progresses through compound formation and reaction phases, mimicking how elements combine to form more stable materials. In the compound formation stage, candidate solutions undergo changes analogous to electron gain, loss, or sharing. This behavior is reproduced through a probabilistic rule based on a Gaussian distribution, generating new material compositions as:

$$PTE_{\text{new}}^k = PTE_{r_1}^k \pm e^- \quad \text{with} \quad f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (11)$$

In addition, MGA models chemical reactions by blending selected candidate materials to produce new combinations. These reactions are mathematically described through a weighted mixture:

$$\text{Mat}_{\text{new}} = \frac{\sum_{m=1}^l (p_m \cdot \text{Mat}_m)}{\sum_{m=1}^l p_m} \quad (12)$$

where p_m follows a normal distribution, imitating the tendency of materials to interact selectively.

Although MGA was originally formulated to solve single-objective problems, many engineering systems require balancing multiple goals simultaneously. This need has led to the development of the multi-objective MGA, which extends MGA to optimize conflicting objectives while maintaining solution diversity. MOMGA incorporates three additional mechanisms [57]: (1) a storage archive for non-dominated solutions, (2) a grid-based diversity controller, and (3) a leader selection

strategy.

The archive ensures that only Pareto-optimal candidates are preserved. A newly generated material is admitted only if it is not dominated by an existing member, or if it dominates one currently stored. When the archive becomes full, MOMGA employs a grid partitioning method to identify crowded regions in objective space and remove an overrepresented member so that diversity is not lost. The leader selection mechanism guides the search toward sparse areas of the Pareto front using a probabilistic rule:

$$P_i = \frac{C}{N_i} \quad (13)$$

where N_i is the number of archived solutions in the i -th grid segment and $C > 1$ is a constant. Less populated regions thus have a higher probability of contributing search leaders, ensuring broad exploration of the decision space.

Overall, MOMGA leverages MGA's chemical analogy while incorporating Pareto dominance and diversity preservation, enabling robust convergence to a well-distributed optimal front for multi-objective engineering applications.

4.2. Pareto optimal points

The MOMGA was implemented in MATLAB R2022a to determine the Pareto-optimal points of the proposed system. The primary control parameters were defined as a population size of 300, a maximum number of 500 iterations, and an external archive size of 150 solutions. For the grid-based leader selection mechanism, the parameters were set to $\alpha = 0.1$, $\beta = 4$, $\gamma = 2$, with $n\text{Grid} = 10$ divisions per objective dimension. These parameter values were selected to provide an appropriate balance between exploration of the search space and exploitation of promising regions, thereby improving the stability and robustness of the optimization process. To reduce the impact of stochastic behavior inherent in evolutionary algorithms, the optimization procedure was repeated five independent times, and the final Pareto archive was selected based on the most stable convergence pattern among the runs.

Fig. 8 illustrates the three-dimensional Pareto frontier obtained using the MOMGA optimization algorithm for the geothermal-based tri-generation system. The goal of the optimization was to simultaneously maximize EE and MFR_{H2} , while minimizing PSC. The distribution of the Pareto solutions reveals a clear multidimensional trade-off among the three performance indicators. The 3D surface projections on EE– MFR_{H2} , EE–PSC, and PSC– MFR_{H2} planes highlight physically consistent trends: higher exergy efficiency is achieved at the expense of increased PSC when hydrogen production rates are higher, as evidenced by the curvature of the Pareto cloud and its reflected shape on the EE–PSC plane. Conversely, points with lower PSC tend to shift toward higher EE values but lower MFR_{H2} values, forming a distinguishable boundary on the PSC– MFR_{H2} surface. This suggests that lowering cost requires compromising the additional work ratio α or reducing membrane temperature, both of which tend to decrease hydrogen output. The EE– MFR_{H2} plane exhibits a steep frontier, indicating that improvements in hydrogen mass flow require a notable decline in process efficiency, likely due to deviations from the separator pressures or evaporator temperature. Importantly, the scatter density and smooth continuity of the Pareto front indicate excellent coverage of the optimization space, with MOMGA successfully capturing both global and local trade-offs, rather than converging prematurely to narrow solution clusters. The algorithm provides well-distributed, non-dominated solutions with diverse cost–efficiency balances, making it highly valuable for techno-economic decision-making.

The convergence behavior of the algorithm was evaluated by monitoring the evolution of the non-dominated solution archive throughout the optimization process. The results showed that the quality of the Pareto front improved rapidly during the first approximately 200

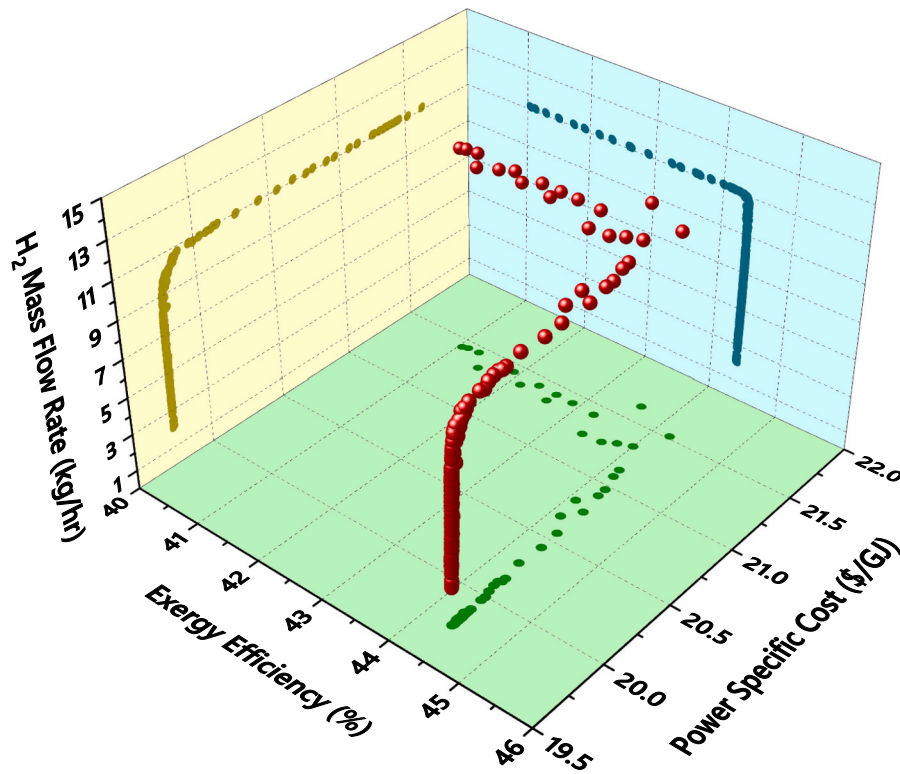


Fig. 8. Pareto-optimal solutions from the MOMGA optimization showing the trade-off among exergy efficiency, hydrogen mass flow rate, and power-specific cost.

iterations, followed by gradual stabilization after about 350 iterations, indicating convergence toward the optimal solution set. In addition, the hypervolume indicator, calculated relative to the reference point ($EE = 40\%$, $PSC = 22$ \$/GJ, $MFR_{H_2} = 1$ kg/h), exhibited a monotonic increase and eventually reached a plateau, further confirming convergence. The diversity of the obtained Pareto front was assessed using the spread

index defined as $\Delta = \frac{d_f + d_i + \sum_{i=1}^{N-1} |d_i - d|}{d_f + d_i + (N-1)d}$, where N denotes the number of

non-dominated solutions, d_i represents the Euclidean distance between two consecutive solutions in the Pareto front, and d is the average of these distances. Moreover, d_f and d_i denote the distances between the extreme solutions of the obtained Pareto front and the corresponding true extreme points. The obtained Δ value of 0.4 indicates a reasonably diverse set of non-dominated solutions with an acceptable distribution along the Pareto front.

Fig. 9 illustrates the distribution of Pareto-optimal input variable values and their corresponding effects on the three performance outputs. For EE, clear plateaus appear in most trends, indicating that once a suitable operating window is selected, increases in the variable beyond a threshold yield marginal improvement. EE values generally peak when P_{S1} and P_{S2} operate at moderate-to-high ranges, where vapor quality entering the expansion and separation units is optimal. Similarly, EE increases with higher T_{EVP} and moderate P_{LNG} , corresponding to improved thermodynamic efficiency through better evaporation conditions and higher ORC circulation pressure. For both α and T_{PEM} , a plateau trend appears at first, but when α exceeds ~ 0.5 and T_{PEM} goes beyond ~ 73 °C, the EE starts to decline, because more work is diverted to electrolysis and higher membrane temperatures increase irreversibilities, reducing overall system efficiency.

In contrast, PSC values exhibit sharp variability when plotted against P_{S1} , P_{S2} , and P_{LNG} , with lower costs associated mostly with lower to mid-range separator pressures and lower pump work requirements. Increasing α , T_{PEM} , and P_{LNG} generally leads to higher PSC values. Higher α allocates more ORC work to the electrolyzer, raising both

energy consumption and the associated capital costs. Elevated T_{PEM} increases the energy required to maintain the PEM at its operating temperature, adding to operational expenses, while higher P_{LNG} increases pump power consumption and contributes to greater energy usage. Conversely, increasing T_{EVP} reduces PSC due to more efficient heat transfer in the ORC, with the cost decreasing until it reaches a plateau at approximately 164 °C, beyond which further increases in T_{EVP} do not significantly affect PSC.

Hydrogen production rate increases with higher P_{S1} , P_{LNG} , α , and T_{PEM} , as these factors improve electrolysis energy availability, ORC output, and ionic conductivity. In contrast, increasing T_{EVP} reduces MFR_{H_2} . For P_{S2} , two distinct trends are observed: one subset of optimal solutions shows MFR_{H_2} decreasing with P_{S2} , while another subset exhibits an increasing trend, indicating competing influences of separator pressure on fluid separation and electrolysis efficiency.

To understand the operational characteristics and trade-offs of the system, the distribution of Pareto points across different input parameters was analyzed to identify predominant operating ranges and their implications for performance, efficiency, and risk. The analysis of P_{S1} shows that most operating conditions, approximately 67% of the Pareto points, fall within the low-pressure range of 603–620 kPa. This suggests that the system predominantly operates under baseline conditions, providing moderate exergy efficiency and hydrogen production. A smaller fraction, around 6%, corresponds to the medium-pressure range of 620–785 kPa, indicating slightly elevated operation where hydrogen production improves marginally, accompanied by a small increase in power-specific cost. The remaining 27% of the points are in the high-pressure range of 785–941 kPa, representing high-to-extreme operational conditions that maximize exergy efficiency and hydrogen output but entail greater economic and mechanical stress. These findings highlight a system tendency toward moderate pressures, with fewer configurations exploring the benefits and risks of higher-pressure operation.

P_{S2} exhibits a different distribution pattern, with 36% of the data in the low-pressure range of 84–92 kPa, reflecting stable baseline operation

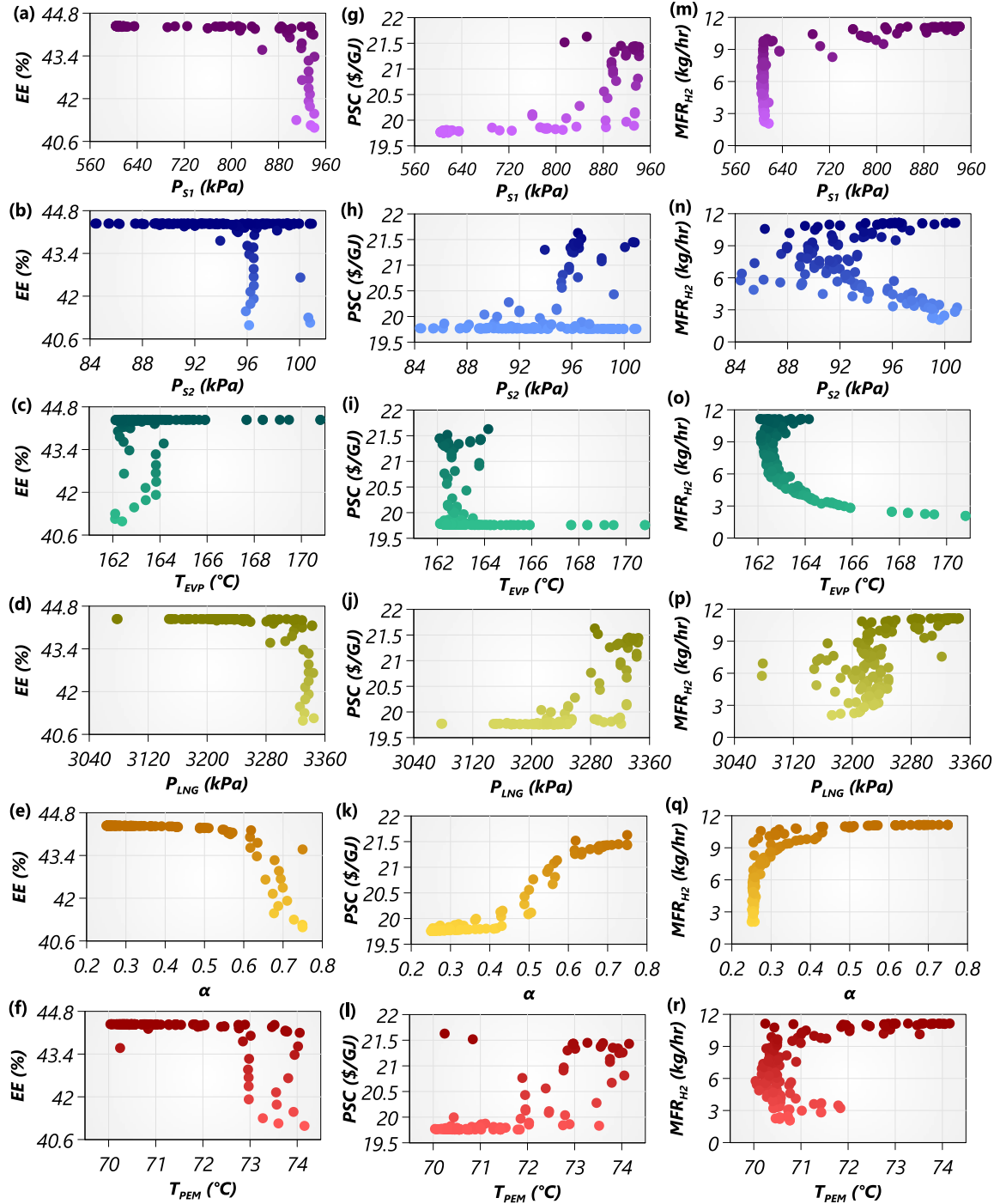


Fig. 9. Pareto-optimal distributions of input variables and their impact on EE, PSC, and MFR_{H2}.

with moderate hydrogen separation efficiency. Approximately 41% of the points lie within the medium-pressure range of 92–97 kPa, indicating elevated pressures that enhance hydrogen recovery at the cost of slightly increased power consumption. The remaining 23% fall in the high-pressure range of 97–101 kPa, corresponding to operation that maximizes hydrogen purity and recovery but introduces higher mechanical stress and operational risk. T_{EVP} is predominantly concentrated in the low-temperature range of 162–164 °C, accounting for 82% of Pareto points. This range ensures stable evaporation, moderate hydrogen separation efficiency, and safe operation. The medium-temperature range of 164–166 °C comprises 13% of the data, representing slightly elevated operation that improves vaporization rate and

hydrogen recovery with a modest increase in energy consumption. High-temperature conditions of 166–171 °C, covering only 5% of the points, correspond to maximal hydrogen output and separation efficiency but introduce greater thermal stress and operational challenges. For P_{LNG} , 51% of Pareto points are in the medium-pressure range of 3220–3320 kPa, reflecting elevated operation that enhances LNG flow rate and system throughput, albeit with slightly higher energy consumption and mechanical load. Low-pressure operation, in the range of 3077–3220 kPa, represents 34% of the points and is associated with safe pumping, moderate LNG flow, and stable energy consumption. The remaining 15% of points fall within the high-pressure range of 3320–3345 kPa, representing conditions that maximize LNG delivery and overall

performance but increase mechanical stress and operational risk. α reveals that 55% of Pareto points occur in the low-share range of 0.25–0.28, indicating efficient operation with moderate hydrogen production as the electrolyzer consumes only a small portion of ORC work. About 25% of the data are in the medium-share range of 0.28–0.45, where the electrolyzer consumes a moderate portion of ORC work, increasing hydrogen output at the cost of slightly reduced power efficiency. The high-share range of 0.45–0.75, covering 20% of points, corresponds to operation dominated by the electrolyzer, maximizing hydrogen production while significantly reducing system exergy efficiency and increasing operational demands. Finally, T_{PEM} data indicate that most Pareto points, approximately 67%, lie within the low-temperature range of 70–71 °C, reflecting safe membrane operation and stable hydrogen production with moderate efficiency. Medium temperatures of 71–73 °C account for 23% of points, representing slightly elevated operation that increases hydrogen production and reaction rates with a modest increase in membrane stress. High-temperature operation of 73–74.2 °C comprises 10% of the data, maximizing hydrogen output and reaction efficiency but increasing the risk of membrane degradation and operational challenges.

To further clarify the origin of the observed optimal operating windows, several mechanistic interactions among the main decision variables and the thermodynamic subsystems can be identified. The additional work ratio (α) directly governs the fraction of ORC power diverted to the electrolyzer. Increasing α enhances the electrical energy available for electrolysis and therefore increases the hydrogen production rate. However, beyond a moderate range, this diversion reduces the net power available to the rest of the system and shifts the energy balance toward electrochemical conversion rather than thermodynamic power recovery. Consequently, the exergy efficiency begins to decline, which explains the appearance of an optimal α window where hydrogen production improves without excessively degrading system efficiency. A similar mechanism explains the influence of the evaporator temperature. Increasing T_{EVP} raises the mean temperature of heat addition in the ORC and improves the thermodynamic potential for work generation. This initially enhances turbine output and reduces the relative exergy destruction within the evaporator due to improved temperature matching with the geothermal heat source. However, once T_{EVP} exceeds a certain level, the marginal increase in cycle work becomes small while temperature gradients in downstream components increase, leading to higher irreversibilities in heat transfer processes. As a result, the improvement in EE and PSC gradually reaches a plateau. The LNG pump pressure primarily affects the cold-energy recovery process and the thermodynamic state of the natural-gas expansion stage. Increasing P_{LNG} raises the pressure level of the LNG stream before vaporization, which increases the potential expansion work recovered in the natural gas turbine and improves cold-energy utilization. At the same time, higher pumping pressure inevitably increases the required pump work and the associated exergy consumption. The competition between these two effects creates a moderate pressure window where cold-energy recovery and expansion work are maximized without excessive pumping losses.

It should be noted that the present optimization framework is based on steady-state thermodynamic modeling and system-level performance indicators. Therefore, although the trends observed in Fig. 9 allow qualitative interpretation of the mechanisms responsible for the optimal operating windows, detailed state-point exergy destruction analysis of each component was not included in the current study. Such component-level analysis could provide further quantitative insight into the irreversibility distribution within the ORC, LNG, and electrolysis subsystems and may be considered as a potential extension of this work.

5. Multi-criteria decision-making

To effectively identify the most desirable operating conditions in a geothermal-based tri-generation system, MCDM methods are essential. While Pareto-optimal solutions obtained from multi-objective

optimization provide valuable insight into the trade-offs between key performance indicators, they do not directly indicate which combination of operating parameters is most advantageous for practical implementation. The system under consideration integrates multiple complex components. Each of the inputs affects system performance in distinct ways, often creating conflicting objectives: for instance, increasing hydrogen production may reduce overall exergy efficiency or elevate operational stresses. In real-world applications, such as hybrid energy systems that recover LNG cold energy while simultaneously producing electricity, natural gas, and hydrogen, selecting the optimal operating point requires a careful balance between maximizing performance and ensuring operational reliability. MCDM provides a structured framework to address this challenge by consolidating multiple conflicting outputs into a single comprehensive ranking system. In this study, WASPAS method is employed to systematically evaluate and rank the Pareto-optimal configurations, facilitating the identification of the operating conditions that most effectively harmonize exergy efficiency, economic feasibility, and hydrogen production. This approach ensures that the selected configuration is not only mathematically optimal but also robust, practical, and suitable for real-world implementation.

5.1. WASPAS technique

The WASPAS method offers a powerful and flexible framework for multi-criteria decision-making, making it particularly well-suited to complex engineering optimization problems [58]. Its main advantage is its capacity to manage multiple, sometimes conflicting objectives within intricate design spaces. A key characteristic of WASPAS is its integration of two complementary evaluation approaches [59]: the weighted sum model (WSM) and the weighted product model (WPM). The WSM evaluates alternatives by summing performance across all criteria, providing a comprehensive overview, whereas the WPM penalizes options that perform poorly in any single criterion, ensuring that significant weaknesses are not concealed by strong results in other areas.

The procedure for applying WASPAS follows a structured sequence of steps. Initially, a decision matrix is constructed to organize the performance of m alternatives across n criteria:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \quad (14)$$

where each element x_{ij} represents the performance of the i -th alternative with respect to the j -th criterion. To allow meaningful comparisons across different criteria, normalization is performed. For criteria where higher values are preferable (beneficial criteria), normalization is given by:

$$\dot{x}_{ij} = \frac{x_{ij}}{\max(x_{ij})} \quad (15)$$

For criteria where lower values are preferable (non-beneficial criteria):

$$\dot{x}_{ij} = \frac{\min(x_{ij})}{x_{ij}} \quad (16)$$

Following normalization, the weighted sum score for each alternative is calculated as:

$$Q_i^{(1)} = \sum_{j=1}^n w_j \dot{x}_{ij} \quad (17)$$

and the weighted product score is computed as:

$$Q_i^{(2)} = \prod_{j=1}^n \left(\dot{x}_{ij} \right)^{w_j} \quad (18)$$

These two scores are then combined into a single overall performance index using a parameter λ ($0 \leq \lambda \leq 1$):

$$Q_i = \lambda Q_i^{(1)} + (1 - \lambda) Q_i^{(2)} \quad (19)$$

Adjusting λ allows the decision-maker to emphasize either a balanced evaluation ($\lambda = 0.5$) or favor one component (WSM or WPM) over the other. Finally, all alternatives are ranked based on their composite Q_i values, with higher scores representing superior trade-offs among system objectives. This ranking facilitates the selection of the most practically efficient and optimized configuration for the system under study.

5.2. Design points

Table 3 shows the one-criteria, two-criteria, and three-criteria decision-making results obtained from the WASPAS technique for a geothermal-based tri-generation system. By applying different weight combinations to the three main objectives (EE, PSC, and MFR_{H2}) various design scenarios (A–J) were generated. These scenarios, presented in Fig. 10 across the Pareto front, provide a spectrum of operational strategies that guide system design, optimize trade-offs between conflicting objectives, and allow decision-makers to adapt to different real-world priorities.

According to Table 3 and Fig. 10, scenarios A, B, and C correspond to cases in which only a single objective is considered. Scenario A represents a design fully optimized for exergy efficiency, where the WASPAS weight is assigned entirely to EE ($W_{EE} = 1$), ignoring PSC and hydrogen production. The optimal inputs for this case include a moderate Separator 1 pressure (603.9 kPa) and a relatively low Separator 2 pressure (84.4 kPa), along with a moderate evaporator temperature (163.4 °C) and LNG pump pressure (3077.2 kPa). α is low at 0.253, and T_{PEM} is 70.038 °C. This configuration achieves an EE of 44.378% while maintaining a modest PSC (\$19.770/GJ) and hydrogen production rate (5.753 kg/h). The rationale for this scenario is to illustrate the maximum exergy efficiency that can be achieved without consideration of cost or hydrogen output, which can be crucial in applications where reducing energy losses is prioritized, such as district heating integration or remote off-grid power systems. Scenario B focuses exclusively on minimizing PSC, with the weighting fully allocated to cost ($W_{PSC} = 1$). The system operates at slightly higher pressures and temperatures than Scenario A ($P_{S1} = 616.7$ kPa, $P_{S2} = 99.6$ kPa, $T_{EVP} = 170.8$ °C, $P_{LNG} = 3172.4$ kPa) but maintains a low α of 0.250. This configuration results in a comparable EE of 44.374% but significantly reduces the cost metric, achieving \$19.757/GJ, albeit at the expense of hydrogen production, which drops to 2.069 kg/h. This scenario is particularly relevant for industrial-scale deployment where capital and operational expenditures dominate project feasibility, and hydrogen generation is secondary to overall system affordability. Scenario C emphasizes maximizing hydrogen production ($W_{MFR} = 1$). This requires operating the separators at much higher pressures ($P_{S1} = 934.6$ kPa, $P_{S2} = 100.8$ kPa) and a higher

electrolyzer work fraction ($\alpha = 0.749$). As a result, the system achieves a notable hydrogen output of 11.143 kg/h while also enhancing EE to 41.129% and slightly reducing PSC to \$21.443/GJ. This scenario highlights the trade-offs when hydrogen yield is prioritized, making it suitable for applications in industrial hydrogen supply or for hydrogen-fueling infrastructure where production capacity is the key criterion, and exergy efficiency and cost are secondary considerations.

According to Table 3 and Fig. 10, Scenarios D, E, and F correspond to two-criteria decision-making cases, offering balanced trade-off solutions between pairs of objectives. Scenario D ($W_{EE} = 1$, $W_{PSC} = 1$) mirrors Scenario B in inputs and outputs, as minimizing cost and maximizing EE align closely. Scenario E ($W_{EE} = 1$, $W_{MFR} = 1$) balances exergy efficiency and hydrogen production, using $P_{S1} = 933.7$ kPa, $P_{S2} = 94.8$ kPa, $T_{EVP} = 162.4$ °C, $P_{LNG} = 3328.8$ kPa, and $\alpha = 0.432$. This configuration yields an EE of 44.335%, a PSC of \$20.153/GJ, and a high hydrogen rate of 10.975 kg/h. Such a scenario is ideal for applications in combined energy-hydrogen systems, such as industrial clusters that require both electricity and hydrogen simultaneously. Scenario F ($W_{PSC} = 1$, $W_{MFR} = 1$) optimizes cost and hydrogen production, producing EE = 44.363%, PSC = \$19.863/GJ, and MFR_{H2} = 10.573 kg/h, with moderate pressures and $\alpha = 0.274$. This scenario suits hydrogen-intensive energy systems with strict cost constraints, such as distributed hydrogen generation units integrated into gas networks.

According to Table 3 and Fig. 10, Scenarios G to J correspond to three-criteria decision-making cases. Scenario G represents a full three-criteria optimization with equal weighting ($W_{EE} = W_{PSC} = W_{MFR} = 1$). The operating conditions mirror Scenario F, achieving a compromise between EE, PSC, and hydrogen output. This balanced scenario offers a versatile baseline for system deployment, allowing designers to meet moderate efficiency, cost, and hydrogen production simultaneously, which is particularly valuable in multi-purpose energy facilities or hybrid power plants where no single objective dominates. Scenarios H, I, and J illustrate the flexibility of the weighting approach in WASPAS. Scenario H ($W_{EE} = 3$, $W_{PSC} = 1$, $W_{MFR} = 1$) emphasizes EE while maintaining moderate attention to cost and hydrogen production, suitable for applications requiring maximum energy utilization with manageable cost and hydrogen production. Scenario I ($W_{EE} = 1$, $W_{PSC} = 3$, $W_{MFR} = 1$) prioritizes cost minimization while balancing the other objectives, ideal for large-scale commercial plants with limited budget but reasonable energy and hydrogen performance. Scenario J ($W_{EE} = 1$, $W_{PSC} = 1$, $W_{MFR} = 3$) focuses on maximizing hydrogen production while maintaining acceptable EE and PSC, directly applicable to hydrogen-focused projects or refueling stations where exergy efficiency and cost are secondary concerns.

Taken together, these scenarios not only support the selection of optimal system designs but also offer actionable insights for real-world applications ranging from low-cost electricity supply to hydrogen-focused industrial energy hubs. The diversity of scenarios ensures that designers have a structured methodology to navigate trade-offs, maximize system performance, and adapt to varying operational

Table 3

WASPAS-based one-, two-, and three-criteria decision-making results for the geothermal tri-generation system, showing design inputs, outputs, and corresponding weights for scenarios A–J.

Scenario	Weight			Input						Target		
	W_{EE}	W_{PSC}	W_{MFR}	P_{S1} (kPa)	P_{S2} (kPa)	T_{EVP} (°C)	P_{LNG} (kPa)	α	T_{PEM} (°C)	EE (%)	PSC (\$/GJ)	MFR _{H2} (kg/h)
A	1	0	0	603.9	84.4	163.4	3077.2	0.253	70.038	44.378	19.770	5.753
B	0	1	0	616.7	99.6	170.8	3172.4	0.250	70.764	44.374	19.757	2.069
C	0	0	1	934.6	100.8	162.1	3344.8	0.749	73.602	41.129	21.443	11.143
D	1	1	0	616.7	99.6	170.8	3172.4	0.250	70.764	44.374	19.757	2.069
E	1	0	1	933.7	94.8	162.4	3328.8	0.432	71.958	44.335	20.153	10.975
F	0	1	1	884.6	86.3	162.5	3254.4	0.274	72.893	44.363	19.863	10.573
G	1	1	1	884.6	86.3	162.5	3254.4	0.274	72.893	44.363	19.863	10.573
H	3	1	1	884.6	86.3	162.5	3254.4	0.274	72.893	44.363	19.863	10.573
I	1	3	1	614.0	91.4	162.8	3237.1	0.417	71.280	44.352	19.810	9.985
J	1	1	3	919.3	93.6	163.1	3244.9	0.362	71.854	44.354	19.969	10.818

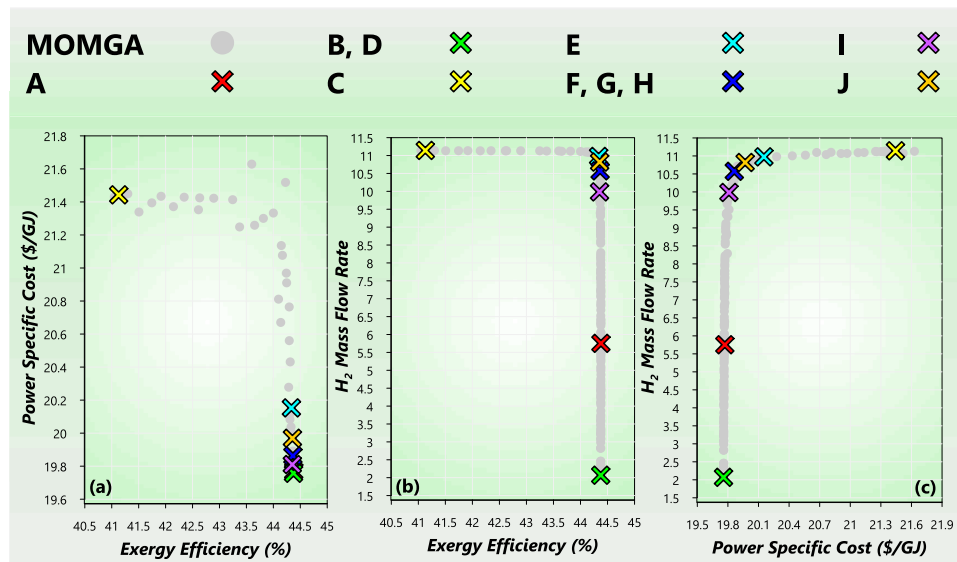


Fig. 10. Pareto front of the geothermal tri-generation system with WASPAS-selected scenarios (A–J).

requirements, thereby enhancing the practical value and robustness of the proposed tri-generation system.

It should be noted that the weight factors used in the WASPAS analysis are preference-dependent. The ten scenarios introduced in this study are intended to represent a spectrum of plausible design priorities that may be adopted based on project-specific considerations. In practical engineering applications, designers may determine the weights through multiple valid approaches, such as structured expert elicitation, surveys of domain specialists, regulatory requirements, or project-specific performance targets. Consequently, no universally optimal solution is claimed; rather, the final design point is selected from the Pareto front once the relevant weighting preferences have been defined.

6. Conclusion

6.1. Study overview

This study presents an advanced intelligent framework for geothermal–LNG assisted tri-generation systems, integrating machine learning, multi-objective optimization, and scenario-based decision-making to enhance system performance. Hybrid MLPNN models optimized via PSO, MPA, and WSO accurately predict key system outputs, capturing nonlinear thermodynamic interactions. MOMGA is employed for multi-objective optimization to explore trade-offs among exergy efficiency, power-specific cost, and hydrogen production, generating a robust Pareto-optimal solution set. The WASPAS method is subsequently applied for decision-making, ranking the solutions to identify practically implementable configurations. This integrated approach offers a holistic methodology for predicting, optimizing, and selecting optimal operational strategies, thereby advancing the design and operation of tri-generation energy systems under complex and conflicting objectives.

6.2. Key findings

Key findings of the proposed framework on the considered data are summarized below:

- **Machine Learning:** The PSO-MLPNN achieved the highest prediction accuracy for EE ($R > 0.999$, $\text{MAPE} < 0.03\%$), MPA-MLPNN excelled for PSC ($R > 0.999$, $\text{MAPE} < 0.02\%$), and WSO-MLPNN performed best for hydrogen production ($R > 0.999$, MAPE

$< 0.3\%$), demonstrating that optimizer selection critically affects predictive reliability for different objectives.

- **Trade-offs:** MOMGA generated a well-distributed Pareto front, highlighting clear trade-offs: higher EE generally increases PSC, while maximizing hydrogen output reduces EE. Input analysis showed the system predominantly favors baseline-to-moderate operating conditions with limited configurations exploring extreme settings.
- **Operational Insights:** Key operating windows were identified: P_{S1} (603–620 kPa), P_{S2} (84–97 kPa), T_{EVP} (162–164 °C), P_{LNG} (3077–3320 kPa), α (0.25–0.28), and T_{PEM} (70–71 °C), ensuring balanced efficiency, hydrogen yield, and cost.
- **System Behavior:** Multi-dimensional trends revealed that improving one objective often requires compromises in others, emphasizing the need for a systematic approach to navigate performance, cost, and hydrogen production simultaneously.
- **MCDM:** WASPAS scenarios demonstrated flexible decision-making: single-objective scenarios prioritized EE, PSC, or hydrogen, two-objective scenarios balanced pairs of objectives, and three-objective scenarios achieved an equitable compromise, providing actionable insights for real-world tri-generation system design.

6.3. Limitations and future directions

Despite its contributions, this study has several limitations. First, the ML models were trained on simulated datasets; experimental validation under real operational conditions is required to confirm predictive robustness. Second, the optimization framework assumes steady-state operation, neglecting dynamic responses and transient behaviors that may occur in practice. Third, uncertainties in LNG supply, ambient conditions, and component degradation were not incorporated, which could impact system performance under realistic scenarios. For future research, integrating real-time data-driven updates to ML models, considering stochastic optimization under uncertainty, and exploring dynamic multi-objective optimization are recommended. Additionally, coupling the framework with techno-economic and environmental assessments could provide a more comprehensive evaluation for sustainable tri-generation deployment. Extending the methodology to hybrid renewable systems may further enhance applicability and decision-making flexibility for multi-energy infrastructure planning.

CRediT authorship contribution statement

Bhishma Karki: Supervision, Project administration, Formal analysis, Data curation, Conceptualization. **Dessalegn Ahmed Yeshanew:** Software, Methodology, Investigation. **Ilkhom Khaydarov:** Visualization, Software, Formal analysis. **Hüseyin Kurt:** Writing – review & editing, Writing – original draft, Software. **Paul Rodrigues:** Software, Formal analysis, Data curation. **Shivam P. Chaudhary:** Writing – original draft, Software. **Shahbaz Juneja:** Writing – review & editing, Resources. **Akanksha Mishra:** Writing – original draft, Software. **Jajneswar Nanda:** Writing – review & editing, Writing – original draft. **Aseel Smerat:** Writing – original draft, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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